



2019 K-Innovation ODA Program with Vietnam

Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology and Innovation (STI) Strategic Planning in Vietnam 2019-2020

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Institute	Science and Technology Policy Institute (STEP I)
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Program Investigator	Wangdong Kim, Chief Director, Division of Global Innovation Strategy, STEP I
Project Manager	Jungwon Lee, Senior Research Fellow, Office of Development Cooperation Research(IICC), STEP I
Researchers	Jungwon Lee, Senior Research Fellow, Office of Development Cooperation Research(IICC), STEP I Byeongwon Park, Research Fellow, Office of Multilateral Cooperation Research, STEP I Yong In Choi, Researcher, Division of Global Innovation Strategy, STEP I Young Il Park, Professor, Ewha University Yoo Hyung Won, Principal Researcher, KIST Hyun Yim, Senior Research Fellow, KISTEP Bach Tan Sinh, Head of Division of Journal and Publication, Department of Science Administration and Training, VISTI. Le Dinh Tien, Former Deputy Minister, Ministry of Science and Technology, Vietnam Ta Doan Trinh, Former President, NISTPASS
Project Coordinator	Yong In Choi, Researcher, Division of Global Innovation Strategy, STEP I

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Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology, and Innovation(STI) Strategic Planning in Vietnam 2019-2020

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5th~7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si, 30147.

Republic of Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit

iicc.stepi.re.kr

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Tel: +82-2-815-0407 Fax: +82-2-822-1269

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Executive Summary

The Vietnamese government has emphasized science, technology, and innovation (STI) as the most important keys for implementing the SocioEconomic Development Strategy (SEDS) 2011-2020 plan and realizing sustainable growth. Vietnam has relatively good STI capacity and human resources among ASEAN countries. Unfortunately, however, Vietnam's STI capacity is still lower than that of other developed countries. In this regard, the Vietnamese government needs a strategic approach to STI policies based on scientific methodologies such as technology foresight, given the limited resources and capacity.

Based on the confirmation and acceptance of the project proposal submitted by NISTPASS (currently VISTI), the Korean government affirmed the support for Vietnam in making faster progress toward achieving Vietnam's STI development goals through the K-Innovation ODA program "Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology, and Innovation (STI) Strategic Planning in Vietnam 2019-2020."

The Project aims to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the strategic planning for Vietnam's STI. The scope of the Project has been defined based on Vietnam's STI policy priorities and the lessons learned from the Korean experiences in national STI policies and technology foresight. Accordingly, the Project framework includes the following: i) review of Vietnam's national STI strategies and technology foresight experiences; ii) research on the Korean experiences in technology foresight and national STI strategies; iii) local workshops for the diagnosis of Vietnam's technology foresight capacities and for the consensus on the policy consultation's direction and contents; iv) capacity building workshop with technical training programs to enhance the technology foresight capacity of Vietnamese government officials and experts; and v) development of the implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts.

At the kick-off meeting with VISTI, the objectives and scope of the Project were discussed. STEPI and VISTI discussed further specific action plans and agreed on the project programs and activities. During the workshop on national future strategy and technology foresight, the researchers of both counterparts shared insights on Korea's S&T strategic planning and its implications on Vietnam's capacity for national STI policy and technology foresight. Based on the lessons learned from the workshop and the consensus on the Project activities, VISTI and Vietnamese experts prepare the draft framework of the pilot exercise, which will be specified and enriched through the capacity building workshop.

The research team suggested implications and recommendations on how to develop Vietnam's capacity for national STI strategy and technology foresight. Regarding the national STI strategy, it is necessary to make the government establish and implement an STI strategy systematically with active commitment to making STI contribute to national socioeconomic development. Vietnam needs to promote strategic investments and systematic management to achieve the best possible outcomes given the limited R&D resources. Vietnam also needs to improve its R&D structure to increase the performance of the R&D system considering the limited R&D resources. For this, the role of R&D organizations such as VAST, VISTI, and VKIST should be clearly identified.

Regarding the technology foresight, Vietnam lacks experts in technology foresight and has a very weak system of government cooperation. There is also a need to develop a technology foresight methodology suitable for the Vietnamese context. The most important factor for the successful implementation and use of technology foresight is recognizing the importance of STI and technology foresight in the political decision-making process and cooperation of all the relevant government ministries.

Vietnam should explore its own growth path through the technology foresight. In this regard, it is important to develop Vietnam's own technology foresight methodology by applying the implications from the Korean experience to the Vietnamese environment and characteristics.

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CHAPTER 1

Project Overview

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation
(STI) Strategic Planning in Vietnam 2019-2020



Chapter 1. Project Overview

1. Introduction

The Vietnamese government plans to become one of the industrialized countries by establishing “Socioeconomic Development Strategy (SEDS) 2011-2020” aiming at political and social stability, improvement of quality of life, and improvement of international status.

The Vietnamese government has historically emphasized the importance of science and technology. In particular, science, technology, and innovation are recognized as the most important keys to implementing the SEDS 2011-2020 plan and realizing sustainable growth. In this regard, Vietnam has promoted science, technology, and innovation (STI) development in order to achieve sustainable socioeconomic development goals.

For the transition to the industrialized economy, technological development is the most important policy. In addition, to achieve the goal of increasing the proportion of hi-tech industries to 45%, S&T capacity for industries such as electronics, ICT, and advanced machinery should be developed.

Vietnam has relatively good STI capacity and human resources among ASEAN countries. Unfortunately, Vietnam’s STI capacity is still lower than that of other developed countries. According to the report of the World Economic Forum, Vietnam ranks 92nd and 73rd in the areas of technology acceptance and innovation, respectively.

In Vietnam, where STI capacity and R&D investment are not sufficient to lead economic and social development, there is a need for a strategic approach to promoting STI policies.

Vietnam’s R&D investment as a percentage of GDP is less than 0.5% (in 2013). This is

significantly lower than that of Malaysia (about 1 ~ 1.25% during the same period) and still lower than Thailand (same period, about 0.3 ~ 0.45%/about 0.6% in 2015). Increasing the effectiveness of the STI policy given such limited R&D investment necessitates a “selection and concentration strategy.” In other words, a national technology strategy is needed, and technology foresight is a prerequisite.

Based on the confirmation and acceptance of the project proposal submitted by NISTPASS (currently known as VISTI), the K-Innovation ODA Program has affirmed the commitment to supporting Vietnam in making faster progress toward achieving its STI development goals.

At the same time, the Korean government designated the official development assistance priority areas for Vietnam based on Korea’s New Policy for South and Vietnam’s development strategies and development needs as well as Korea’s comparative advantage.

Accordingly, the scope of the Project shall be based on the consideration of Vietnam’s STI policy priorities and the lessons learned from the previous Korean experiences in national STI policies and technology foresight.

The aim of the Project is to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Vietnam.

The Science and Technology Policy Institute (STEPI) is a Korean government-sponsored research institute and a leading global think tank that conducts research and analysis on issues pertaining to science, technology, and innovation. STEPI works closely with various ministries, regional governments, and relevant institutes to support policy design. It also has a long experience in providing STI policy consultancy for developing countries, having the role of conducting STI development cooperation activities with partner countries.

2. Objectives

The project is titled “Policy Consultation on Strengthening the Capacity for Foresight for Science, Technology, and Innovation (STI) Strategic Planning in Vietnam 2019-2020.” The program duration is from January 1, 2019 to December 31, 2019 (12 months).

- 1) The objective of the K-Innovation ODA Project with Vietnam (2019-2020) is to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Vietnam.
- 2) The detailed objectives are as follows:
 - Review the current status of STI policy and technology foresight in Vietnam
 - Share Korea’s experience in technology foresight
 - Conduct a capacity-building workshop
 - Provide technical advice on the draft report of pilot-scale exercise of a Strategic Plan on Technology Foresight

3. Project Framework

The Project's main activities will be framed as follows:

- Review of Vietnam's national STI strategies and technology foresight experiences
- Research on Korean experiences in technology foresight and establishment of national STI strategies or plan based on technology foresight
- Local workshops for the diagnosis of Vietnam's technology foresight capacities and matching work and for the consensus on the direction and contents of policy consultation for major policymakers, government officials, and research institutes of Vietnam
- Capacity-building workshop to enhance the technology foresight capacity of Vietnam's government officials and experts in the STI field through technical training programs
- Development of implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts

[Figure 1-1] Conceptual Framework of the Project



Source: author

4. Project Team

4.1 Research Team

Based on the Project's purposes, a research team was composed with STEPI researchers and three experts. The experts are invited to share their knowledge of Korean development experiences.

The research team consists of three researchers, including Project Manager (PM) Dr. Jungwon Lee and two STI policy experts in STEPI. The research team is responsible for the review on Vietnam's STI strategic plan and technical advice on technology foresight exercise. During the training courses and consultation sessions, the team will provide constructive comments on the pilot exercises of Vietnam and share their evaluation on the Project's progress. External experts have participated in the Project to provide lectures during the workshop and to share technical information or materials for the topics covered.

[Table 1-1] Research Team

Name	Organization	Area of expertise	Contact information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	STI Policy, Technology Foresight	leejw@stepi.re.kr
Byeongwon Park	STEPI, Research Fellow	Technology Foresight	bwonpark@stepi.re.kr
Yong In Choi	STEPI, Researcher	Project Coordinator	luciachoi@stepi.re.kr
Young Il Park	Ewha University, Professor	Strategic Planning	yipark@ewha.ac.kr
Yoo Hyung Won	KIST, Principal Researcher	STI Policy	yhwon@kist.re.kr
Hyun Yim	KISTEP, Senior Research Fellow	Technology Foresight	hyim@kistep.re.kr

4.2 Local Research Team

As one of the main institutions of Vietnam in the STI field, VISTI, as a policy-oriented research institute, was designated as a counterpart of STEPI for conducting the entire process of the Project. Aside from a researcher from VISTI, Dr. Bach Tan Sinh, two more additional local experts were invited to provide in-depth analysis and related background knowledge of Vietnam's STI policy and Technology Foresight exercises.

[Table 1-2] Local Research Team

Name	Office	Position	Contact information
Bach Tan Sinh	VISTI	Head of Division of Journal and Publication, Department of Science Administration and Training, VISTI	sinhbt@gmail.com
Ta Doan Trinh	-	Former President of NISTPASS	tadoantrinh1956@gmail.com
Le Dinh Tien	-	Former Deputy Minister of Science and Technology	haletien@yahoo.com.vn

5. Project Activities

5.1 Activity 1: Preparation of the Project

From January to March 2019, the STEPI team conducts a literature review related to the STI strategic planning and technology foresight of Vietnam. Any relevant documents on the issue to enhance understanding should be provided before Activity 2. Detailed agenda for the kick-off meeting and first workshop will be developed based on the Project Concept Proposal officially received from the Ministry of Planning and Investment of Vietnam and shared with VISTI for coordination.

5.2 Activity 2: Kick-off Meeting and First Workshop

The STEPI team visits Hanoi, Vietnam on April 15 to 19, 2019 for the kick-off meeting and the first workshop. Specifically, the workshop aims to share Korea's experiences in the identified topics, including national strategic planning for STI, introduction of technology foresight, and strategic planning of Government Research Institutes. The Vietnamese counterpart will give a presentation on the STI Policy of Vietnam and technology foresight in Vietnam in order to promote in-depth understanding of Vietnam's current status and experiences in foresight. Afterward, STEPI and VISTI will build consensus on the expected results and outcome.

5.3 Activity 3: Capacity-building Workshop in Korea

The capacity-building workshop will be held in October 2019 to share the best practices of Korea and to practice technology foresight as an implementation strategy of the STI Policy. VISTI shall invite a delegation of ten (more or less) national experts on the STI policy, and STEPI shall manage the workshop activities. STEPI will provide guidelines of the capacity-building workshop for one week. For the pilot-scale exercise of a Strategic Plan on Technology Foresight, VISTI shall deliver its interest in key areas during the workshop.

5.4 Activity 4: Review on a Draft Report of Pilot-scale Exercise of a Strategic Plan on Technology Foresight

A draft of the report on the pilot-scale exercise of a strategic plan on Technology Foresight will be reviewed by the research team and Korean experts for progress assessment.

5.5 Activity 5: Final Workshop

A final workshop will be held in November or December 2019. During the workshop, VISTI will present the final report on a Strategic Plan on Vietnam's Technology Foresight for the final comments from the STEPI team as well as Vietnamese stakeholders from the public and private sectors, academe, and research institutes involved in the STI policy. STEPI will present the results and achievement of the cooperative project with Vietnam and follow-up plan for 2020. Additionally, an evaluation interview can be conducted to assess the relevance and sustainability of recommendations and to build upon a possible follow-up of the Project.

6. Project Schedule

The entire schedule of the project is as follows:

[Table 1-3] Project Schedule

Activities	Jan. '19	Feb. '19	Mar. '19	Apr. '19	May '19	Jun. '19	Jul. '19	Aug. '19	Sep. '19	Oct. '19	Nov. '19	Dec. '19
Project Identification												
Pre-study & Literature Review												
Kick-off Meeting and Workshop												
STI System and TF Review												
Interim Report												
Capacity-building Workshop												
Comments on Progress												
Final Dissemination Workshop												
Final Report												

7. Project Outputs

STEPI and VISTI will draw up the following deliverables:

- **Deliverable 1:** Project Proposal (STEPI)
- **Deliverable 2:** Program Agenda for Kick-off Meeting and Workshop (STEPI)
- **Deliverable 3:** Program Agenda for Capacity-building Workshop in Korea (STEPI)
- **Deliverable 4:** Presentations and Training Materials for Capacity-building Workshop (STEPI)
- **Deliverable 5:** Draft of Pilot-scale Exercise of a Strategic Plan on Technology Foresight (VISTI)
- **Deliverable 6:** Final Report of a Strategic Plan on Technology Foresight (VISTI)



CHAPTER 2

STI Policy and Technology Foresight in Vietnam

Policy Consultation on Strengthening the Capacity of
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(STI) Strategic Planning in Vietnam 2019-2020

Chapter 2. STI Policy and Technology Foresight in Vietnam

1. Vietnam's STI Status

1.1 Overview¹

Vietnam has had a remarkable development record over the past 30 years. Economic and political reforms under Đổi Mới, launched in 1986, have spurred rapid economic growth and development and transformed Vietnam from one of the world's poorest nations to a lower middle-income country.

Vietnam's economy continues to show fundamental strength, supported by robust domestic demand and export-oriented manufacturing. The extreme poverty rate is estimated to have declined to below 3 percent. Following the 6.8 percent growth in 2017, preliminary data indicate that GDP growth soared to 7.1 percent in 2018, underpinned by a broad-based pickup in economic activity.

Vietnam's medium-term outlook is broadly favorable, and downside risks are tied to weak external demand, shifting trade patterns, global financial volatility, and incomplete banking and state-owned enterprise (SOE) reforms. On the upside, Vietnam is strongly positioned to benefit from numerous free trade agreements that are coming into force now and over the forecast period.

Vietnam's growth is projected to be a moderate 6.6 percent in 2019, driven by credit tightening, slower private consumption, and weaker external demand. Inflationary

¹ Source: JST (2014); CSIRO (2019); World Bank in Vietnam, <https://www.worldbank.org/en/country/vietnam/overview#1>, accessed 17 September 2019.

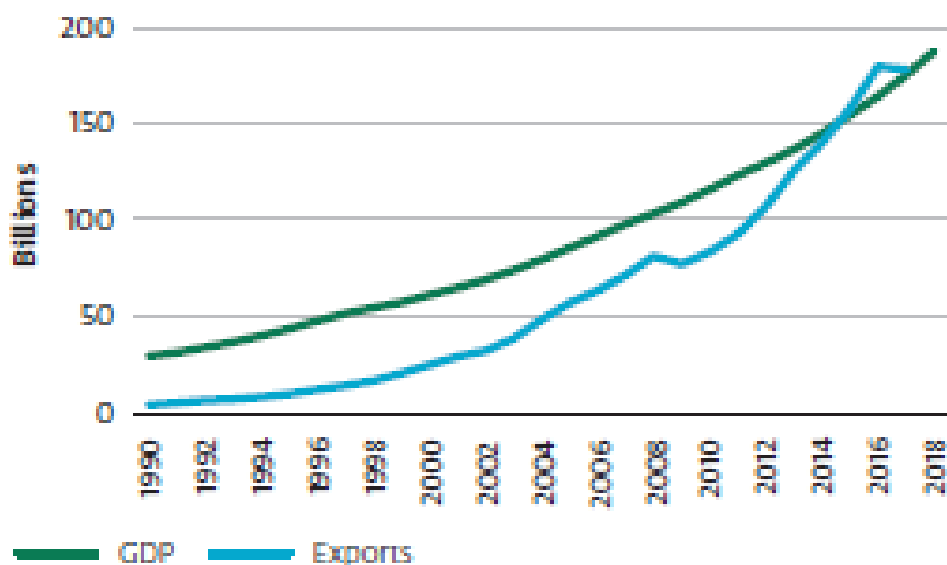
pressures are seen to remain moderate, due to subdued global demand conditions and moderate global energy and food prices. Over the medium term, growth is projected to stay at around 6.5 percent as the impact of the current cyclical uptick dissipates. Poverty is expected to decline further as labor market conditions remain favorable.

In Vietnam, science and technology are considered to be a driving force for the development of a modern industrial society. Legislation, planning, and organizations related to science and technology have been developed, with policies implemented in line with the “Science and Technology Development Strategy (2011-2020).”

According to the World Bank, Vietnam’s scientific and technological potential has been enhanced. Investments from the state budget for science and technology had an average increase of 16.5%/year, accounting for approximately 2% of the total state budget expenditure. Social investment in science and technology accelerated, reaching 1.3% of GDP in 2015 (WB, NDP2016-2020).

Nonetheless, research and development expenditures as a percentage of GDP are low at 0.21% (2011), and government research institutes are spending about 60% of the total research funds. Therefore, strengthening research and development in universities and enterprises is a future challenge.

[Figure 2-1] Vietnam's GDP and Exports (constant 2020 US\$), 1990-2018



Source: CSIRO (2018)

Some STI-related indices have shown a more comparable data of Vietnam's STI status.

[Table 2-1] Comparison of Asian Countries in terms of Global Innovation Index

	Overall ranking		Innovation input sub-index		Innovation output sub-index		Innovation efficiency ratio	
	2015	2018	2015	2018	2015	2018	2015	2018
Brunei	-	67	-	37	-	112	-	124
Cambodia	91	98	96	103	91	84	80	60
Indonesia	97	85	114	90	85	73	42	66
Malaysia	32	35	31	34	34	39	56	48
Myanmar	138	-	139	-	130	-	75	-
Philippines	83	73	101	82	77	68	44	62
Singapore	7	5	1	1	20	15	100	63
Thailand	55	44	62	52	50	45	43	33
Vietnam	52	45	78	65	39	41	9	16
S. Korea	14	12	15	14	11	12	27	20

Note: Data for 2015 and 2018 include 141 and 126 countries, respectively.

Source: WIPO (2018); WIPO (2015); edited by author.

1.2 Science and Technology Policy²

Vietnam's S&T and innovation policy objectives are still formulated in the context of a planning approach to defining broader socioeconomic goals.

Aligned with the “Strategy for accelerating industrialization and modernization with a socialist orientation for basically becoming an industrialized country by 2020” (2001-10) and “National Socioeconomic Development Plan 2016-2020,” The Vietnamese government announced the “Science and Technology Development Strategy for 2011-20.”

“Science and Technology Development Strategy for 2011-20”

In April 2012, the Prime Minister of Vietnam approved the Science and Technology Development Strategy for 2011-20. The general goals of the Socioeconomic Development Strategy (2011-2020) are to become a modern industrial country by 2020 and stabilize the political community, improve people's lives, protect the country's sovereignty, and achieve advanced status in the global market. The goals and main targets are as follows:

Goals

- To harmonize humanities and social sciences, natural science, and technological development and make science and technology a driving force for the development of a modern industrial society.
- To reach ASEAN/global level in some areas by 2020.

Targets

- To raise the value of high-technology and applied science products to about 45% of the GDP by 2020 and ensure 15-17% annual growth in the science and technology market.

² OECD (2014); JST (2014)

- In order to increase the ratio of scientific researchers and professional staff in science and technology to 9 or 10 per 10,000 people, up to 5,000 highly skilled engineers will be trained to operate in Vietnam's spearhead fields by 2015.
- To establish 60 basic and applied science research centers of international standing by 2020.

[Table 2-2] Quantitative Goals of S&T Strategy

Goals	2015	2020
High-tech products as a percentage of GDP		45%
Science and technology expenditures as a percentage of GDP	1.5%	2%
Government budget for science and technology in total budget	More than 2%	More than 2%
Number of researchers and engineers per 10,000 people	9-10	11-12
Development of high-tech engineers	5,000	10,000
Number of international-level research institutions	30	60,
Number of science and technology companies	3,000	5,000
High-tech incubators	30	60

Source: JST (2014)

The political and economic culture of Vietnam strongly encourages sectoral rather than thematic prioritization, although the latter has attracted more attention in recent years notably in the context of the Science and Technology Development Strategy for 2011-2020.

The sectoral prioritization set in the S&T Development Strategy for 2011-2020 includes ICT, Biotechnology, Advanced materials, Electronics and semiconductor, Manufacturing mechanics and automation, and Spatial technologies.

The Vietnam Academy of Science and Technology and the Vietnam Academy of Social Sciences will play a leading role in implementing the strategy, but the development of S&T activities by the business sector will also be encouraged. Regarding central public governance, S&T management will gradually shift from a state-owned model to self-reliant mechanisms in accordance with the law.

1.3 R&D³

1.3.1 R&D expenditure (and budget for S&T)

(In terms of R&D Expenditure) According to a survey conducted by the Vietnamese government in 2012, the total R&D expenditures in Vietnam in 2011 stood at 5,294 billion dong (about 26.5 billion yen), and the ratio against GDP was 0.21%. This is considerably low compared to that of Japan, lower than that of Singapore and Malaysia, at the same level as that of Thailand, and higher than that of Indonesia and the Philippines among ASEAN countries. It has not changed much from the 2002 figure (2002: 0.19% Source: UNESCO SCIENCE REPORT 2010).

(Note) With regard to the total research and development expenditures in Vietnam, science and technology statistics have previously been inadequate.

In 2012, a statistical survey was conducted in compliance with the OECD method under the direction of the prime minister. The results are published in “Science and Technology in Vietnam 2013.” The abovementioned total research and development expenditures are lower than the science and technology budget of the government in 2011. There are concerns that the figure for total research and development expenditures is not adequate.

In fact, NISTPASS of MOST independently shows a trial calculation of about 0.8% for the ratio of total research and development expenditures against GDP. Personnel of MOST and NASATI provided the following explanation for this point in the survey:

- The “science and technology budget” has wider use than “research and development expenditures.”
- The capital investment for research and that for production cannot be divided. The science and technology budget includes capital investment, but the other does not. Since investigating this point is beyond the coverage of this survey, the officially published figures were used as they are.

³ OECD (2014); JST (2014)

The following are the characteristics of Vietnam: the activities of (i) government research institutions are active, but those of (ii) universities and (iii) companies are stagnating. The comparison in the percentage of using the research and development budget between Japan and Vietnam is as follows, although there is some difficulty in comparing:

[Table 2-3] Percentage of Using the Research and Development Budget (%)

	Government	University	Company
Japan (2012)	7.9	20.6	71.5
Vietnam (2011)	58.1	14.4	27.4

Source: JST (2014)

(In terms of S&T Budget) The science and technology budget of the Vietnamese government is shown in Table 2-4. Although the amount is increasing significantly, the budget of the whole government is also increasing, and the ratio of the science and technology budget to the total budget is decreasing slightly. A target of more than 2% as a percentage of the total budget for science and technology is set in the 10-Year Science and Technology Strategy, which has not been achieved yet.

[Table 2-4] Budget for Science and Technology in Vietnam

(Unit: Trillion dong)

Year	Budget for S&T	Increase rate	Percentage in total budget (%)
2006	5.429		1.85
2007	6.310	16.22	1.81
2008	6.585	4.36	1.69
2009	7.867	19.46	1.62
2010	9.178	16.66	1.60
2011	11.499	25.28	1.58
2012	13.168	14.51	1.46
2013	14.144	7.41	1.44

Source: JST (2014)

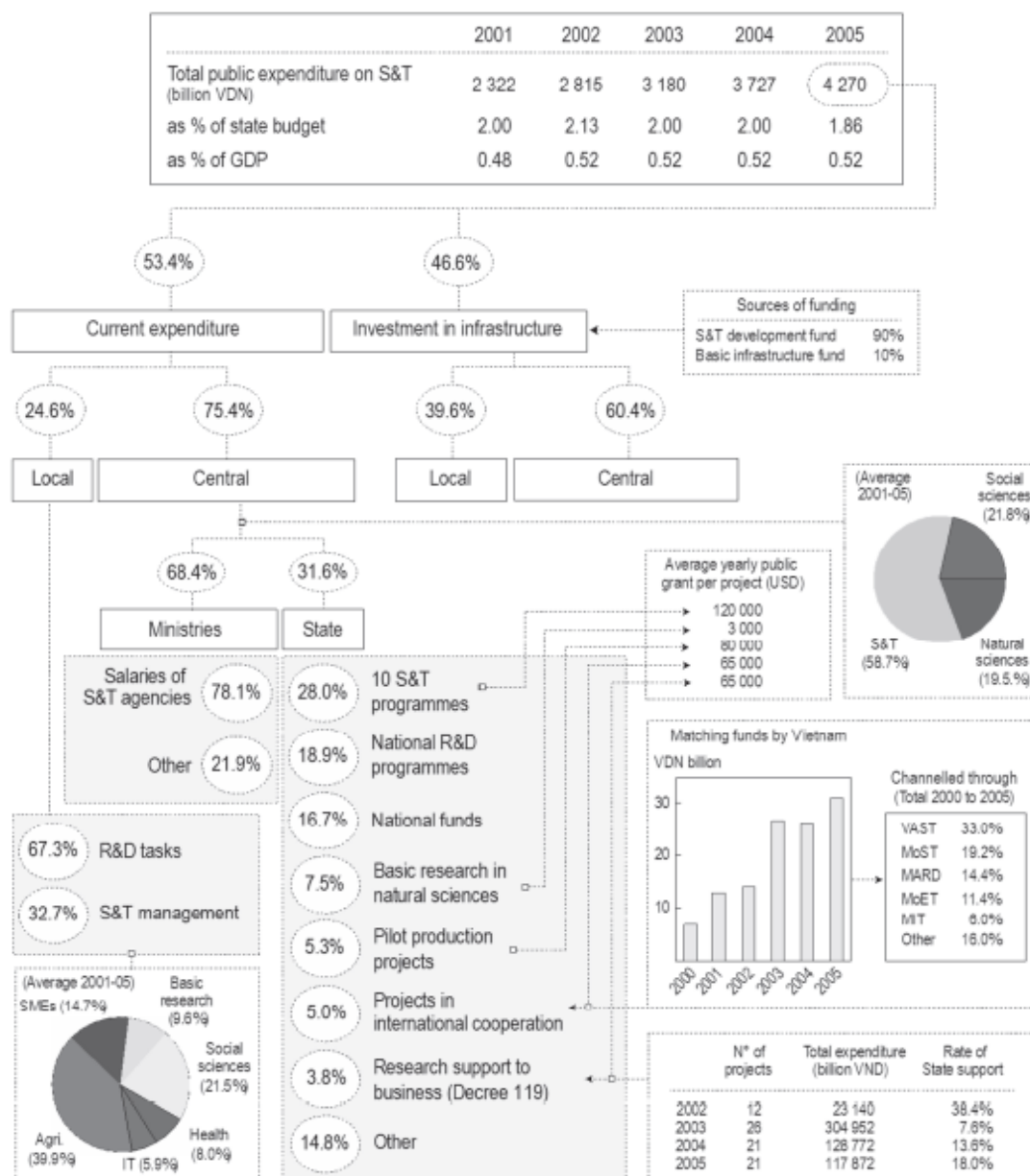
(Mechanism Analysis) STI activities in Vietnam depend heavily on budget support, which represents approximately 65-70% of the total national expenditure on S&T. Public expenditure on S&T grew from approx. 0.2% of GDP in 1995 to around 0.5-0.6% in 2012. More than two-thirds goes to applied research and technological development, and the rest, to basic research.

Even after adjusting for lower labor costs, the available data suggest that many projects are below critical mass and can be justified at best as capacity-building trials. This points to the need for the Vietnamese government to ensure the highest efficiency in the use of limited budgetary means (concentration to ensure impact) while filling major gaps in the system (comprehensiveness to increase systemic efficiency).

Another striking feature is that public expenditure on S&T is mostly channeled through ministries, incurring significant management costs. Although the available data do not allow estimating precisely the amount that ultimately benefits those undertaking research activities, there are concerns that a considerable part of the S&T budget is absorbed by the administrative parts of ministries and beneficiary organizations. Widespread red tape complicates and slows all administrative processes in Vietnam, an issue that is further discussed below.

The main steering mechanisms in Vietnam's STI system are direct control over R&D organizations, conditional provision of public goods (information) and basic infrastructures, financial support to R&D organizations and activities in the form of grants (expenditures) and, albeit marginally, tax incentives (foregone revenues) especially for FDI. Other instruments such as open procurement policies, public-private partnerships (PPPs), and other more demand-side instruments do not play any significant role at present.

[Figure 2-2] Evolution and Allocation of Vietnam's Public Expenditure on S&T



Source: OECD (2014), based on data from NASATI and NISTPASS

1.3.2 Research Output

Based on the benchmark of SciVal (research and analysis tool) developed by Elsevier, the number of science papers, quotes, authors, etc., of papers published between 2009 and 2013 are compared with Japan and 10 ASEAN countries (see Table 2-5). The number of

papers in Vietnam is about the same as that of Indonesia and the Philippines and smaller than that of Malaysia, Singapore, and Thailand among ASEAN countries. The number of citations -- which represents the quality of the papers -- also falls short of that of Singapore.

[Table 2-5] Comparison of Selected Countries by Number of Science Papers

Country \ Item	Number of papers	Number of citations	Number of authors	International Co-authorship
Japan	648,938	3,534,908	599,167	23.6%
Malaysia	93,406	292,001	76,671	31.3%
Singapore	80,680	701,014	48,757	51.5%
Thailand	53,334	257,150	48,585	37.4%
Indonesia	15,728	58,632	17,247	55.0%
Vietnam	12,696	60,540	13,670	67.9%
Philippines	7,354	47,088	7,747	57.3%
Cambodia	1,064	10,905	1,258	88.8%
Brunei	879	2,373	747	48.9%
Laos	750	4,237	810	93.6%
Myanmar	558	1,651	663	60.8%

Source: JSTP (2014); original source from Elsevier, SciVal Database

The table below shows data on patents in Vietnam. It indicates that both application and registration of patents are predominantly made by non-residents (foreigners). This is common to all ASEAN countries. The number of international applications based on the Patent Cooperation Treaty in Vietnam was 18 in 2011. This number is small compared to more than 600 in Singapore.

[Table 2-6] Number of Applications and Registrations at the National Office of Intellectual Property (2012)

	Total numbers	Residents	Non-residents
Application	3,805	382 (10%)	3,423 (90%)
Registration	1,068	52 (5%)	1,016 (95%)

Source: JST (2014); original source from WIPO

1.4 Human Resources⁴

1.4.1 Overall Status

Vietnam currently has one of the lowest numbers of R&D researchers relative to the size of its population. In contrast, in China and Korea, the relative number of R&D researchers per million population more than doubled from 2000 to 2008 (WDI, 2012; OECD, 2014).

[Table 2-7] R&D Researchers (per million population) in East Asia

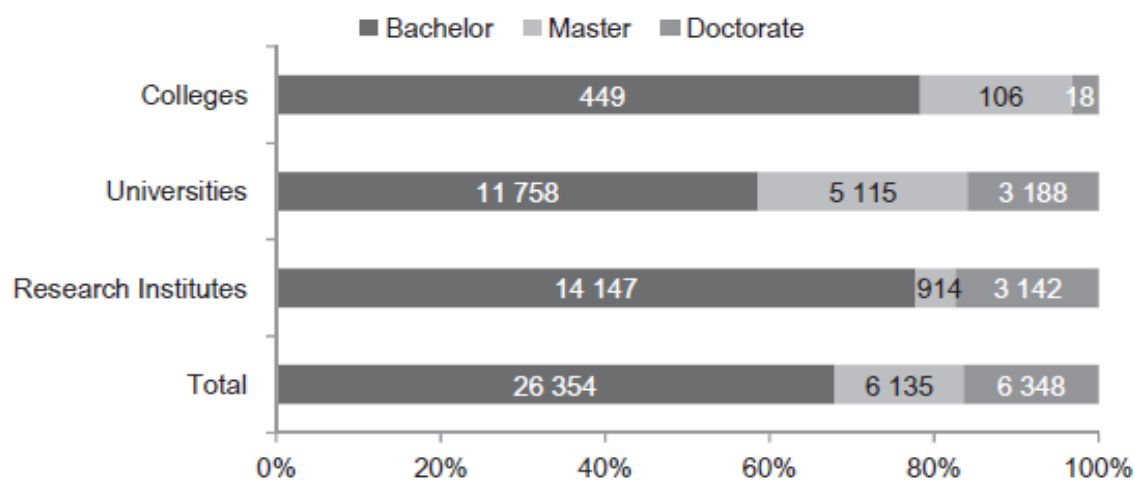
Country	R&D researcher (per million population)
Indonesia	90
Vietnam	116
Thailand	316
Malaysia	365
China	1,199
Korea	4,947

Source: OECD (2014); original source from WDI

Researcher quality also needs to improve. Majority of researchers with tertiary education and in GRIs, particularly in colleges and GRIs, hold only a Bachelor's degree (See Figure 2-3). Universities have a higher percentage of researchers with a Master's degree. Altogether, only 16% of researchers in these institutions hold a doctorate degree, and only 3% of researchers have doctorates in colleges. Nonetheless, Vietnam has an ambitious goal of tripling its science and technology faculty by 2020 with up to 75% of its university lecturers in informatics technology holding a PhD (MoST, 2011).

⁴ OECD (2014)

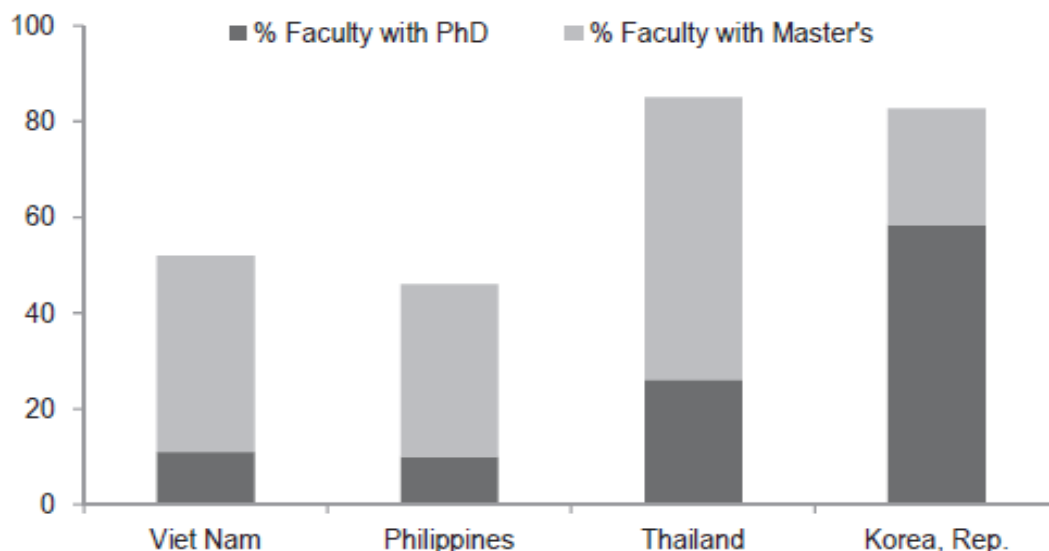
[Figure 2-3] Qualifications for Researcher, by Institution



Source: OECD (2014); original source from NASATI

The major reason is Faculty constraints in universities and TVET system, i.e., insufficient quantity and quality. Given the high and increasing student-faculty ratios, many universities and other tertiary-level institutions in Vietnam have hired insufficiently prepared instructors, and too few of them are PhD holders. For example, about 52% of academic staff in 2010/11 had postgraduate qualifications, and very few had doctorate degrees. Moreover, with the expansion in coverage, the share of faculty with doctorate degrees has decreased from about 15% in 1999/2000 to 11% in 2012.

[Figure 2-4] Ratio of Faculty with Master's and PhD Degrees: Vietnam and Comparator Countries, Various Years



Source: OECD (2014); original from Vietnam: MoET

1.4.2 Human resource gaps in S&T governing bodies (S&T managers)

Although NIS has the institutions and conditions needed to facilitate innovation, it is fragmented and in need of improvement. The system remains weak, isolated, decentralized, bureaucratic, uncoordinated, *ad hoc*, and lacking in clear responsibilities. It includes over 3,200 managers, each of whom takes part in formulating policies and procedures, implementing S&T programs, and supervising staff.

Vietnam's Ministry of Science and Technology (MoST) is the main ministry for coordinating S&T, building capacity, guiding national research programs, and measuring the impact of S&T. Ministries that are also involved in S&T include the Ministry of Industry and Trade (MoIT), Ministry of Agriculture and Rural Development (MARD), and Ministry of Education and Training (MoET). There are provincial Departments of Science and Technology (DoST) offices in all 63 provinces and 1,100 GRIs reporting to various agencies at the central, provincial, and university levels.

The MoST-affiliated State Agency for Technology and Innovation (SATI) and the Vietnamese

Chamber of Commerce and Industry (VCCI) have S&T managers responsible for technology transfer and dissemination at the national, regional, and provincial levels. Other S&T agencies include the Vietnam National Foundation for Science and Technology Development (NAFOSTED), with its own managers at the central level.

In such decentralized and fragmented system, there is no single agency with overarching power; STI governance cuts across a broad, diverse range of public- and private-sector actors. Each agency is relatively isolated, with little coordination or formal power.

In addition to the overall fragmentation of NIS, **its human resource capacity is weak in terms of managing science, technology, and innovation (STI) policies** and specific activities as well as the many GRIs. At all levels, STI managers have limited knowledge and skills for managing the innovation system and promoting technology transfer and dissemination. Therefore, the government has sought to increase the quality and quantity of S&T managers at the central and local levels and within GRIs. Currently, less than 10% of S&T managers have a Master's degree. The government's goal is to increase the number of managers from 3,934 to 4,400 in 2015 (a 12% increase) and to 5,290 in 2020 (a 20% increase). The largest expansion will take place within MoST, where the government aims to double the number of managers within 10 years, and emphasizes increasing the number of managers with a Master's degree from 165 in 2015 to 955 in 2020 (a 479% increase). By 2020, the government aims to increase the number of managers by 50% in other ministries and by nearly 25% at the provincial and local levels.

1.5 Areas for improvement

- Further improve human capital
- Promote industrial development and entrepreneurship
- Reform Government Research Institutions
- Build up networking and open innovation
- Create the framework condition

2. Vietnam's National STI System

2.1 STI Policy Governance⁵

Vietnam's S&T and innovation policy objectives are still formulated in the context of a planning approach to defining broader socioeconomic goals. As shown in Figure 2-5, the long-term socioeconomic development plan is the basis for the formulation of five-year regional and sectoral plans that are then translated into annual plans.

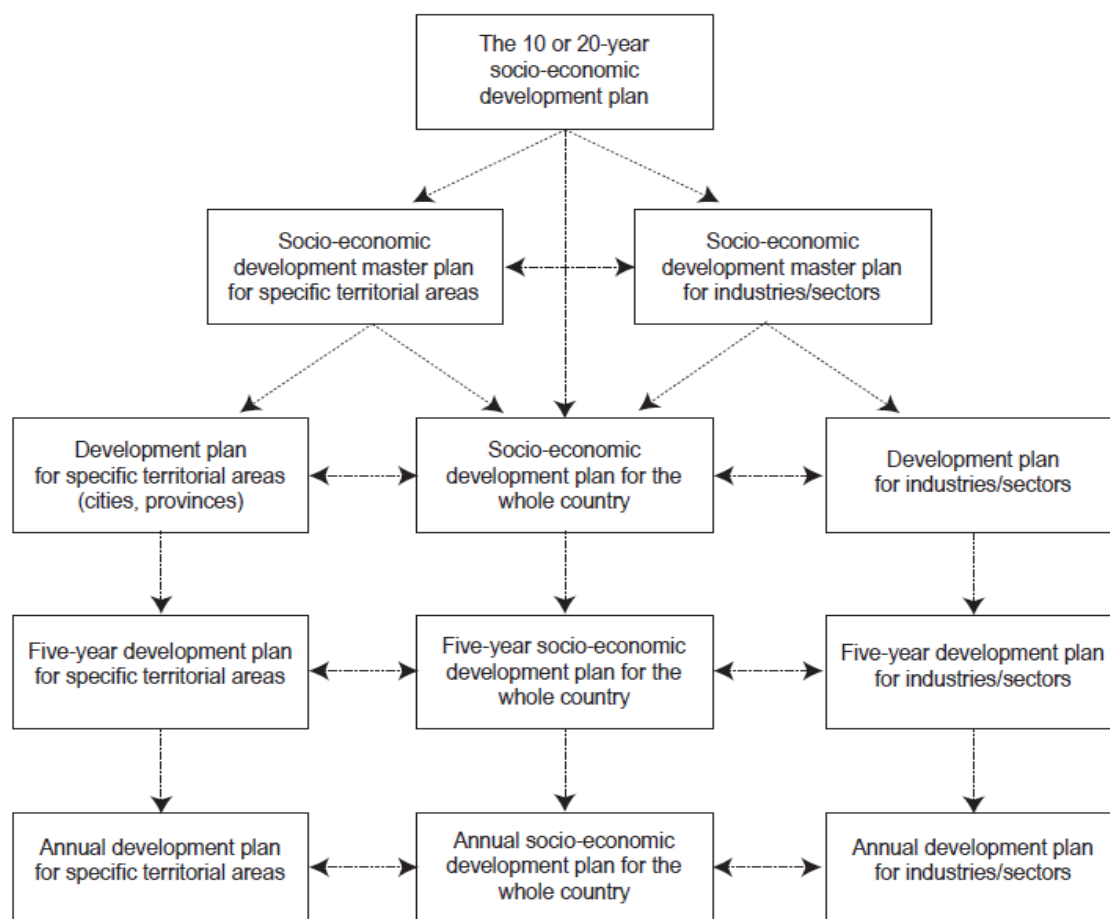
Since 1991, three socioeconomic development strategies (SEDS) have been formulated:

- “Strategy for socioeconomic stabilization and development” (1991-2000);
- “Strategy for accelerating industrialization and modernization with a socialist orientation for basically becoming an industrialized country by 2020” (2001-10);
- “Socioeconomic development strategy 2011-2020”

Based on the SEDS, ministries have to set lower-level strategies and master plans for their respective sectors.

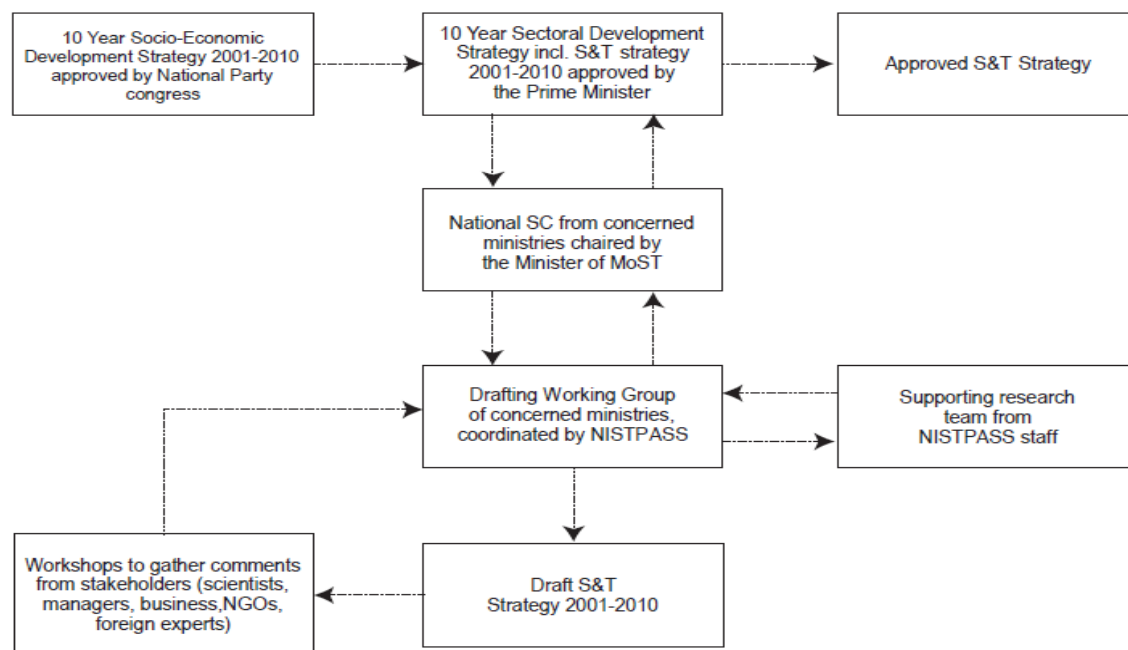
⁵ OECD (2014)

[Figure 2-5] Coordination of Long-term and Short-term Socioeconomic Development Policy



Source: OECD (2014); Huong Lan Nguyen (2011)

[Figure 2-6] Decision-making Process in S&T Policy: Case of Strategy 2010



Source: OECD (2014); original source from Le Dinh Tien (2015)

2.2 Organizations

2.2.1 Government⁶

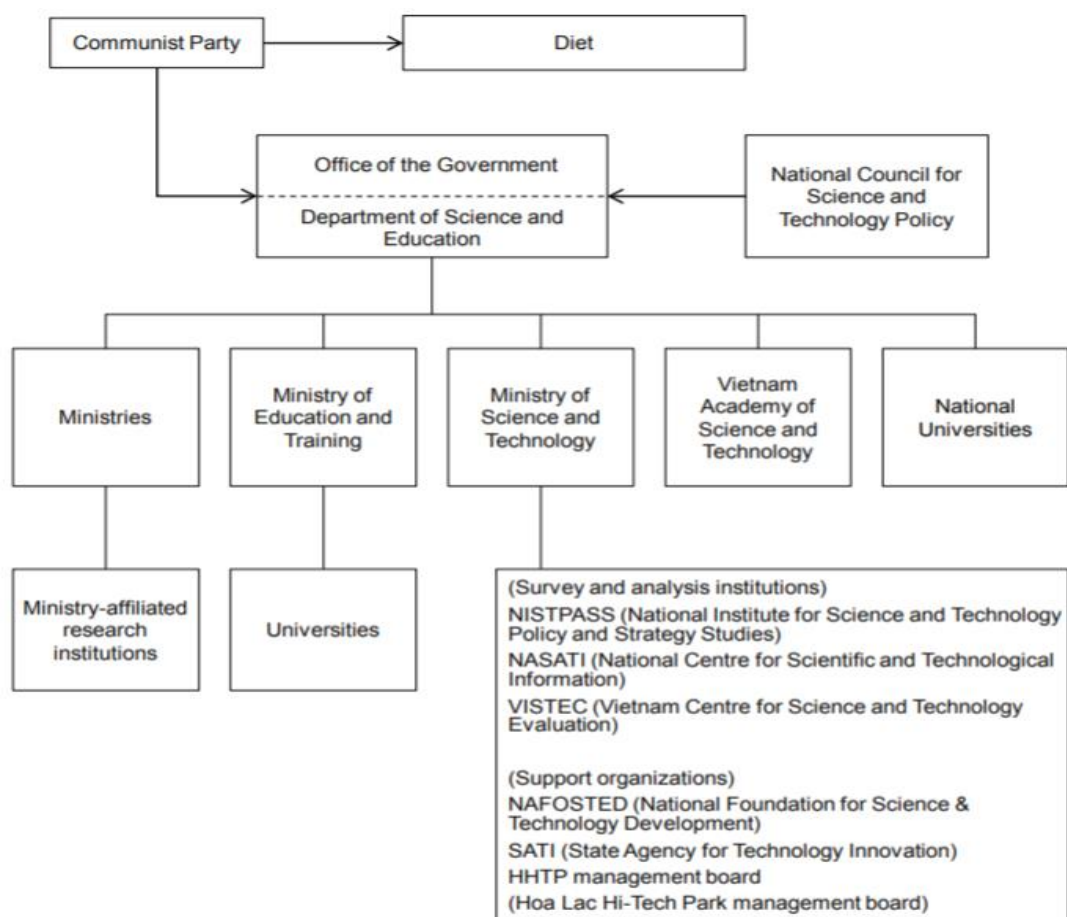
Vietnam's S&T management system consists of:

- Ministry of Science and Technology (a minister is a member of the cabinet)
- Department of Science and Technology in each line ministry
- Department of Science and Technology in each province/city
- 3 levels of S&T management:
 - Level 1. National research institutes or centers directly under the Prime Minister
 - for example, the National Center for Natural Sciences and Technologies and the National Center for Social Sciences and Humanities receive most of their resources from the state budget.

⁶ OECD (2014); KSP (2010)

- Level 2. The research institutes (natural, social, medical, educational, agricultural) and technological branch institutes under ministries, cities, and provinces are also directly under the Prime Minister, receiving most of their resources from the state budget.
- Level 3. The local research units in, for example, districts or cooperatives that are to apply the results supplied by those institutes mentioned above need to meet the local conditions.

[Figure 2-7] Government Organization Chart (central government) related to Science and Technology



Note: NISTPASS is the former name of VISTI (Vietnam Institute of Science, Technology, and Innovation)

Source: OECD (2014)

The National Council for Science and Technology Policy (NCSTP) is positioned as an advisory body to the prime minister. NCSTP consists of 31 members, and the chairman is former

Science and Technology Minister Chau Van Minh. VAST (Vietnam Academy of Science and Technology), VASS (Vietnam Academy of Social Science), and Vietnam National Universities (Hanoi, Ho Chi Minh) are organizations under the direct control of the prime minister.

The Ministry of Science and Technology (MOST) is positioned as the ministry administering science and technology affairs. The Minister, Nguyen Quan (born in 1955, PhD), served as the senior department chair of the Hanoi University of Science and Technology, Vice Minister of Science and Technology in 2007, and Minister of Science and Technology in 2011. MOST administers general science and technology affairs and implements planning and coordination of the entire government. It is also in charge of nuclear power, intellectual property, and standards and metrology. The Ministry of Science and Technology has research, analysis, and planning institutions and research and development support organizations.

Each ministry in the central government has affiliated research institutes that implement activities concerning science and technology. It is important to know the activities of not only the Ministry of Science and Technology but also those of all the other ministries when understanding science and technology in Vietnam. The Ministry of Education and Training (MOET) is in charge of higher education institutions such as universities. There are many research institutes, including the Vietnam Academy of Agricultural Sciences, affiliated with the Ministry of Agriculture and Rural Development. The Ministry of Health and the Ministry of Industry and Trade also have many research institutes.

2.2.2 Institutions⁷

“Agencification,” wherein ministries delegate functions such as research funding to semi-autonomous bodies, is a relatively new phenomenon in Vietnam, where the traditional political culture has always preferred hierarchical to partly contractual arrangements for the delivery of public services. Therefore, the creation in 2003 of the National Foundation for Science and Technology Development (NAFOSTED) was an institutional breakthrough.

⁷ OECD (2014)

The Ministry of Science and Technology (MOST) is in charge of general science and technology affairs as well as nuclear power, intellectual properties, and standards and metrology. The Vietnam Agency for Radiation and Nuclear Safety, Hoa Lac Hi-Tech Park Management Board, National Office of Intellectual Property (NOIP), and Directorate for Standards, Metrology, and Quality are placed under MOST.

Vietnam's main institutions in the STI field can be selected as follows:

Vietnam Institute of Science, Technology, and Innovation (VISTI)

VISTI is a state S&T institution established under the Ministry of Science and Technology; it has full legal status and functions under the automation mechanism of self-reliance, self-responsibility as stipulated by the Government. Originally, it was established as NISTPASS (National Institute for Science and Technology Policy and Strategy Studies) by Decision No. 248/TTg on 23rd April 1996 by the Prime Minister, only to be transformed into VISTI in 2018.

VISTI's key mandates are as follows:

- Research
(conduct theoretical and practical research on STI policy, R&D of advanced technologies; publish scientific research)
- Training
(provide graduate degrees, organize training programs, etc.)
- Support and implement activities related to innovation and startups
(technological incubations, spin-offs and startups, develop and manage co-worker spaces, etc.)
- Policy advisory activities
(participate in the formulation of STI-related laws, strategies, plans, and policies and evaluate the implementation)
- Collaborate with national and international organizations in providing consultancy, training, technology transfer, etc.

Vietnam Academy of Science and Technology (VAST)⁸

The Vietnam Academy of Science and Technology (VAST) is a government agency, previously known as the Vietnam Academy of Science, established in 1975 by Decree 118/CP dated 20 May 1975 and issued by the former Council of the Vietnam Government. The VAST is committed to:

- Carrying out basic research in natural sciences and technology development;
- Providing objective grounds for science and technology management, for shaping policies, strategies, and plans for socioeconomic development;
- Training high-quality human resources for science and technology

VAST has approximately 4,000 employees.

- 2,425 are permanent staff members
- 218 are professors and associate professors
- 840 are doctors and Doctors of Science
- 905 are Masters of Science

In 2016, VAST published a total of 2,000 scientific works. The total number of international articles in 2016 stood at 996 articles, representing a 24.2% increase compared to 2015. The number of articles published in prestigious international journals with ISI standards (SCI and SCI-E) has increased to 742 (which is 26.1% higher compared to 2015), wherein many papers published in journals with high impact factors show that the research quality of VAST has improved a notch. In 2016, VAST funded 69 basic research projects covering 241 subjects. All in all, during the period 2010 - 2016, VAST has received funding for 392 basic research projects from NAFOSTED; this accounts for about 30% of the country's research projects. The total registration of intellectual property has been 28 certificates, including 11 patents and 17 utility solutions (a 56% increase compared to 2016). VAST published 39 monographs

⁸ VAST (2016)

in 2016, a significant increase compared to 2015.

In the field of materials science, “Advances in Natural Sciences: Nanoscience and Nanotechnology,” the journal jointly published by VAST and IOP Publishing (UK), has become Vietnam’s top international standard journal. The journal of the Institute of Mathematics -- Acta Mathematica Vietnamica (AM V) -- is indexed by Scimago, and its ranking has increased from Q4 in 2015 to Q2 in 2016. Furthermore, two journals, VJM and AM V, are indexed for the first time in the Emerging Sources Citation Index (ESCI) by ESCI - IP & Science - Thomson Reuters - Web of Science. This is a clear indication of its ongoing improvement in quality over the last three years.

VKIST (Vietnam-Korea Institute of Science and Technology)⁹

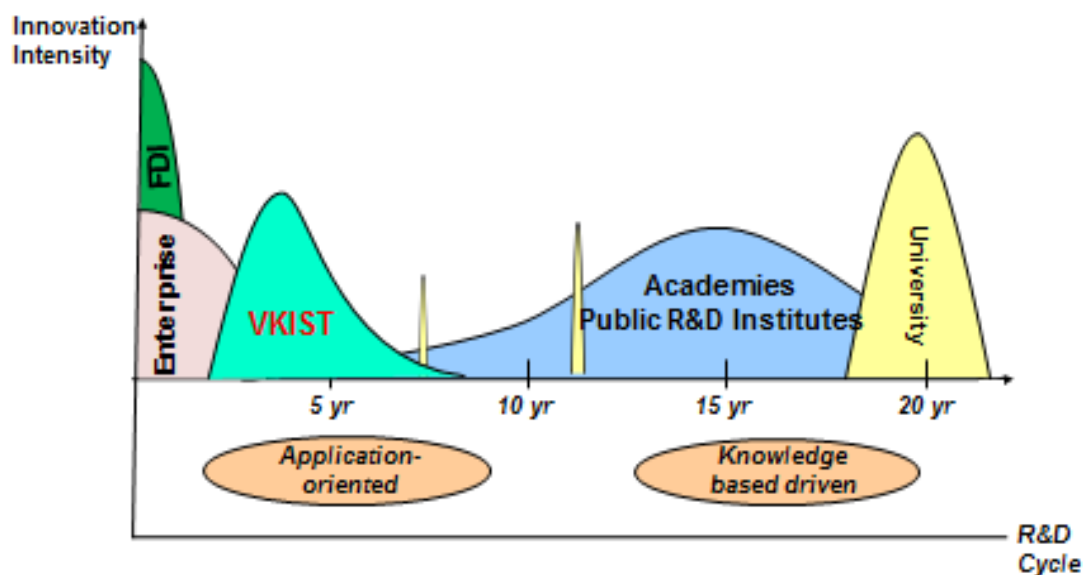
As a new public agency on science and technology under the Ministry of Science and Technology (MOST), VKIST was established through a joint project between the Vietnamese and Korean governments following the Presidential Agreement on September 9, 2013.

Identity of VKIST

- Public science & technology agency under the Ministry of Science and Technology (MOST)
- Research & development (R/D) institute of industrial technologies
- A contract research & development institute
- Accredited agency for full leverage of financial autonomy

⁹ VKIST website, <http://www.vkist.gov.vn/c2/en-US/gioi-thieu-c/About-us-1>, accessed on 14 Oct. 2019.

[Figure 2-8] Position on the National Innovation System (NIS)



Application-oriented R/D of industrial technology with 5 ±a market years

*The vision of VKIST is to become a “Leading technology **solution provider** for market winners in Vietnam.” VKIST has its missions based on Articles 1.3 of the Decree on Establishment of VKIST, No.50/2015/ND-CP;*

- Develop source technologies
- Do research on upgrading technologies for key and strategic industries
- Develop national key products
- Develop integrated technologies to replace imported technologies

The R&D portfolio has been organized based on the missions above:

Internal R/D:

- Market survey on focus area and subject field
- Platform-type research and development

R/D projects:

- National R/D projects
- Enterprise R/D projects
- International cooperation

National Foundation for Science and Technology Development (NAFOSTED)

NAFOSTED was established in 2008, and the annual budget is approximately 10 million dollars (1.1 billion yen). It mainly funds basic research (60% of the budget) but also subsidizes the research activities of small and medium-sized enterprises, fellowship programs, and participation in international conferences. The number of subsidies provided for the basic research on natural science was about 1,100 for six years from 2009 to 2014.

National Agency for Science and Technology Information (NASATI)

The National Agency for Scientific and Technological Information (NASATI, formerly NACESTI), under the Ministry of Science and Technology (MOST), was established on 24 September 1990 based on the merging two organizations: the Central Library on Science and Technology (founded in 1960) and the Central Institute for Scientific and Technical Information (CISTI) (founded in 1972). Acting as the focal point in the national network of S&T information service organizations, NASATI has a mandate of carrying out S&T information activities, disseminating and promoting S&T achievements, organizing and managing Techmarts, and acting as a Central Library for S&T and national registration of R&D project reports.

3. Vietnam's Technology Foresight

3.1 Introductory events for Technology Foresight in Vietnam¹⁰

a) Hanoi "Symposium on Technology Foresight for Development"

- Date and venue: 23 April 2001, Hanoi
- Hosted and sponsored by the Ministry of Science, Technology, and Environment Vietnam
- Co-organized by the APEC Technology Foresight Center and the National Institute for S&T Policy and Strategy Studies (NISTPASS)
- Scope: Within the framework of the 20th APEC ISTWG meeting in Hanoi
- Objective of the symposium: Dissemination of new approach for setting development priorities in APEC member economies including Vietnam
- Participants:
 - International participants from Australia, Canada, China, Japan, Korea, Malaysia, Thailand, and the US
 - Vietnam
- High-level representatives from major ministries, e.g., Gov Office, MPI, MOF, MOET, MOI, MARD, MoFi, MOC, MOT, MOD, MOSTE, and selected DOSTE
- Representative of research institutions and universities
- Representative of business communities such as large corporations

¹⁰ Bach Tan Sinh (2019); Tran Ngoc Ca (2019)

b) (follow-up workshop) Foresight workshop for Food Processing in Vietnam

- Date and Venue: 22-25/10/2001, Do Son
- Organized by NISTPASS
- Supported by APEC Center of TF
- Participants: 30 from research institutions – MARD and MOSTE
- Aims:
 - Introduction of foresight and scenario-planning techniques
 - Creation of scenarios to highlight issues for the food processing sector in VN
 - Exploration of future and setting up of priorities

c) (follow-up workshop) Foresight workshop for the tea sector in Vietnam

- Date and Venue: 25-27/11/2002, Do Son
- Organized by NISTPASS and VINATEA
- Supported by APEC Center of TF
- Participants: 40 from research institutions / MARDMOSTE and Tea Corporation
- Aims:
 - Introduction of foresight and scenario-planning techniques
 - Creation of scenarios to highlight issues for the tea sector in VN
 - Exploration of future and setting up of priorities

3.2 Major Projects¹¹

Vietnam's technology foresight activities have two parallel directions as follows:

- *Foresight*, to explore the future developments of the Vietnamese STI system, both at the level of the national STI system and for specific key technologies, by combining exploratory and normative approaches and to define concrete options and roadmaps for future action.
- *STI strategy* that will translate the findings of the foresight into a series of “**position papers**” that can easily be fed into the formulation of the STI strategy development process. In turn, insights generated in the context of the STI strategy would be fed back into the foresight exercise as well.

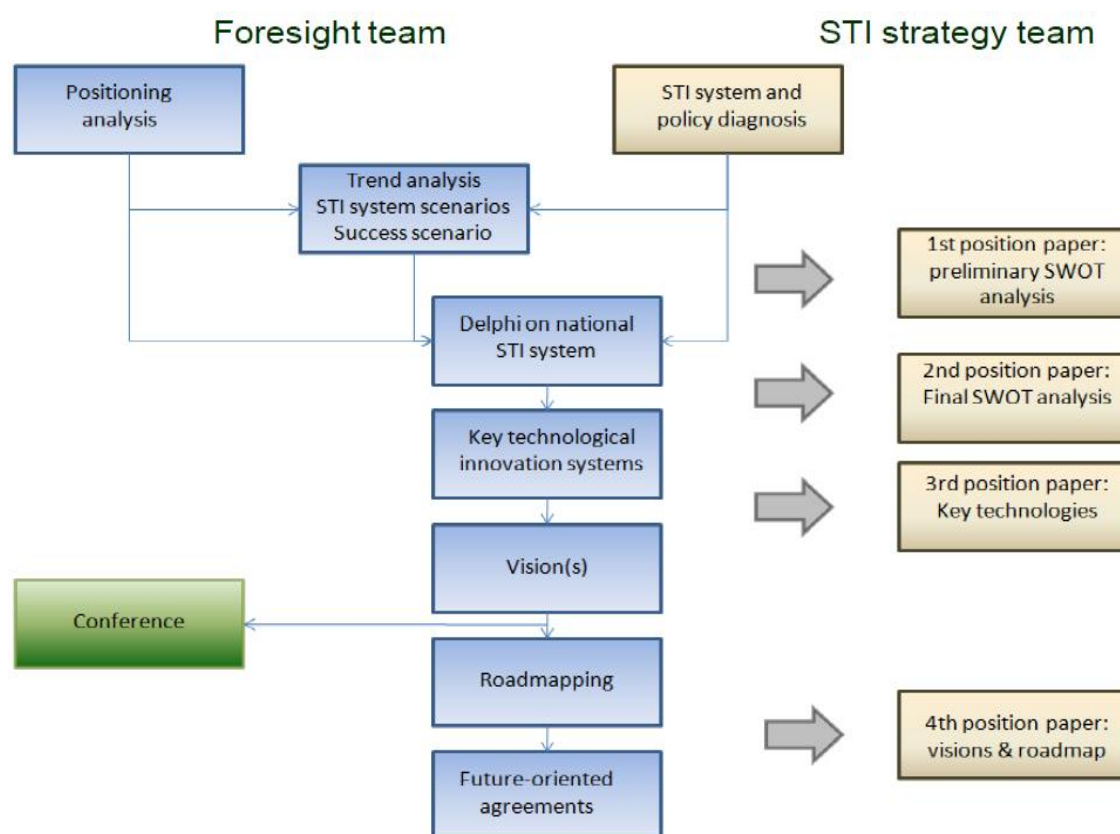
a) Policy Advice for Science, Technology, and Innovation Strategy (STI) 2011 –2020 and High Technology Law Implementation (1)

- Rationale
 - MOST and NCSTP involved in reforming the institutional framework for STI and policies and strategies
 - NISTPASS assigned for S&T policy and strategy formulation 2011-2020, implementation of Law of High Technology approved in 2010
- **Scope:** UNIDO provided technical assistance to NISTPASS in implementing the Project
- Specific aims
 - Strengthen the Vietnamese stakeholders' capacities in designing policies and strategies on science, technology, and industrial innovation to be suitable for the socioeconomic goals of the country.
 - Support the formulation of the 2011 –2020 Strategy by identifying STI policy priorities, i.e., priorities defined along thematic terms (e.g., key science domains,

¹¹ Bach Tan Sinh (2019); Tran Ngoc Ca (2019)

- key technology areas, and key fields of application) or in structural terms (e.g., key cross-cutting deficits in the STI system).
- Support the definition and implementation of priorities, interventions, and future directions of the Vietnamese STI policy and strategy by developing consensual visions of the future through foresight exercises.

[Figure 2-9] UNIDO-MOST Project Structure



Source: Bach Tan Sinh (2019)

b) Integrated Foresight for Sustainable Economic Development and Eco-Resilience in ASEAN Countries, 2013

- Supported by the Rockefeller Foundation
- Implemented by the APEC Center for TF jointly with ASEAN member institutions including NISTPASS
- Outcomes: The successful completion of the project will shape a model from the use of Foresight technique to support sustainable development throughout the ASEAN member states. The results of the project can be used to formulate a new set of insights as to what may be possible by 2020 after the completion of the ASEAN Economic Community in 2015 in terms of stable, productive, and innovative employment prospects in emerging ASEAN economies.

[Table 2-8] Workshops on Technology Foresight with ASEAN

Workshop	Venue	Date	Participants	Expected Output
Workshop 1: ASEAN Context	Sampran Riverside, Nakorn Pathom, Thailand	17-18 Jan 2013	- Approx.30 participants from Thailand (10 representatives each for the 3 issues of energy, water and food - Foresight experts - energy, water and food experts from international organizations	ASEAN's results and prospective foresight for the recommendations of policies, strategies and actions of ASEAN energy, water and food issues.
Workshop 2: Indonesia Local Context	Indonesia	01-02 April 2013	- Approx.30 participants from Indonesia (10 representatives each for the 3 issues of energy, water and food - Foresight experts	Indonesia's results and prospective foresight for the recommendations of policies, strategies and actions of ASEAN energy, water and food issues.
Workshop 3: Vietnam Local Context	Vietnam	04-07 April 2013	- Approx.30 participants from Vietnam (10 representatives each for the 3 issues of energy, water and food) - Foresight experts	Vietnam's results and prospective foresight for the recommendations of policies, strategies and actions of ASEAN energy, water and food issues.
Final Workshop: ASEAN Context	Thailand	01-02 October 2013	- Approx.30 participants from all the 10 ASEAN member states (3 representatives for each 10 ASEAN countries) - Foresight experts - energy, water and food experts from international organizations	- Results conclusions and implications of the previous 3 workshops - Online Real-time Delphi results discussions - ASEAN member states' results and prospective foresight for the recommendations of policies, strategies and actions of ASEAN energy, water and food issues.

Source: Bach Tan Sinh (2019)

c) Digital Economy 2040 Project

- Rationale
 - The Vietnamese government acknowledges that Vietnam, as a fast-growing country in transition, needs to be provident in the face of digital transformation.
- Key Participants
 - Department of Foreign Affairs and Trade Australian Embassy in Vietnam
 - Ministry of Science and Technology State Agency for Technology Innovation
 - Australian Government
 - CSIRO – Data 61
 - DTT Technology Group
- Method
 - Trend Identification: Horizon scanning to identify political, economic, social, technological, legal, and environmental trends
 - Data Analysis: Interviews, Workshops, Economic modeling
 - Futures Analysis: Megatrends driving the digital economy to 2040, Scenarios of Vietnam's digital economy in 2040
- Output
 - Roadmap for Vietnam's Future Digital Economy:
 - Low income – Subsistence farming and urban retail
 - Middle income – Low cost labor + Manufacturing
 - Upper-Middle income – Technology-driven productivity from tech adoption
 - High income – Technology producer and industry leader
 - Recommendation for key areas for future actions:
 - Digital Skills, E-government and procurement, Taxation, ICT and Energy Infrastructure, Cybersecurity and data governance, National Innovation and Industry 4.0

3.3 Brief Analysis of Vietnam's Technology Foresight Capability¹²

- Lack of human resources capable of conducting technology foresight
- Insufficient research on the methodology applicable to Vietnam's socioeconomic conditions
- Weakness of the governance system for technology foresight (e.g., national-level strategy, foresight process, government support, etc.)
- Insufficient high-level strategic directions or guidelines for the technology foresight exercise to be implemented in designing national strategies
- Dispersed training or promotional activities throughout the country
- Reasonably good sign of building up technology foresight capability by collaborating with foreign resources

¹² Tran Ngoc Ca (2019)



CHAPTER 3

Sharing Korea's Experiences in Science and Technology Planning and the Technology Foresight

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation
(STI) Strategic Planning in Vietnam 2019-2020

Chapter 3. Sharing Korea's Experiences in Science and Technology Planning and the Technology Foresight

1. National Strategic Planning for STI Development¹³

1.1 Rationale of the Long-term STI Development Plan

Governments develop long-term STI Development Plans in response to system failure challenges in the STI field. In responding appropriately to the fast speed and great magnitude of science and technology, government-level policy direction greatly impacts the adaptation and utilization of the technological changes. If the market would be the only one handling the changes, a “winner takes all” problem will occur. Therefore, the government should intervene to survive the global competition in science and technology.

There are many other sectors that require government-level plans, such as: Macro and Microeconomic and Budgetary plan, Rural development plan, Energy and resource management plan, Strategic industrial development plan, Transportation and land management plan, Social development plan, etc. Other than these plans, the Science, Technology, and Innovation (STI) plans can be the one that overarches the effectiveness of the other aforementioned plans. The distinguished roles of STI plans are described below:

- Promoting STI development itself, especially through the Strategic Planning of National R&D Programs
- Determining the Development Strategies and related systems
- Leading other sectors' plans by providing forecasting results and practical guidelines and criteria for establishing sectoral plans

¹³ Young Il Park (2019)

- Regulating activities in both STI fields and other sectors

Korea has declared the importance of the national STI plan by legislating the law. Article 7 of the Framework Act of S&T clearly states that “The Government shall set mid- and long-term policy objectives and directions related to the development of science and technology in order to achieve the purpose of this Act efficiently, and shall finalize them following deliberation by the National Science and Technology Council under Article 9 (1).”

The article declares the development of a STI master plan every five years. According to Article 7 of the Framework Act of S&T, “(2) The Minister of Science and ICT shall formulate a master plan for science and technology (hereinafter referred to as "master plan") every five years. It considers the mid- and long-term policy objectives and directions under paragraph (1) and plans, policies, etc. for science and technology formulated by the relevant central administrative agencies. It shall finalize the master plan following deliberation by the National Science and Technology Council in accordance with Article 9 (1).”

1.2 Basic Framework for the STI Development Plan

This section introduces the basic elements and framework of the STI Development Plan, which has been drawn from both general theories and Korean cases.

1.2.1 Characteristics of the STI Development Plan

The establishment of the STI development plan involves various stakeholders from the government, research institutes, private sectors, civil society, etc. Therefore, coordination and consensus among these stakeholders are the key success elements of planning.

In addition, the Plan has to be coordinated with other sectors’ plans and existing institutions. The capability of the government officials to build the plan in coordination with those institutions would be one of the important factors.

Based on the consensus among the stakeholders and the coordination with other

institutions, the plan should be based on future foresight. Future foresight that reflects the dynamics of the socioeconomic changes of society would give broad, forward-looking insights for the planning.

At the same time, the country's development stage should also be considered in order to set a strategic approach. The appropriate sub-goals and action plans for the future goal should be planned based on the current status and capacities of the country.

After setting the goals and action plans, implementation and evaluation should be conducted appropriately. The mid- and final evaluation will give insights for establishing the next long-term STI plans successfully.

1.3 Planning Mechanism and Process

There are four planning mechanisms with which the national STI Development Plan is solidified and rationalized. The level of planning, the key actions, and the key elements will be as follows:

- Cabinet level – coordination
- Ministerial level – Planning and Implementation
 - Sectoral plans, Experts groups, Integrating Body
- Supporting agencies – Foresight and Data Analysis
 - Foresight and forecasting, Trends analysis, Providing data and indicators, Issue monitoring and studies, etc.
- Public Participation – consensus building, monitoring, and evaluation

Based on the planning mechanisms, the planning process can be described as follows:

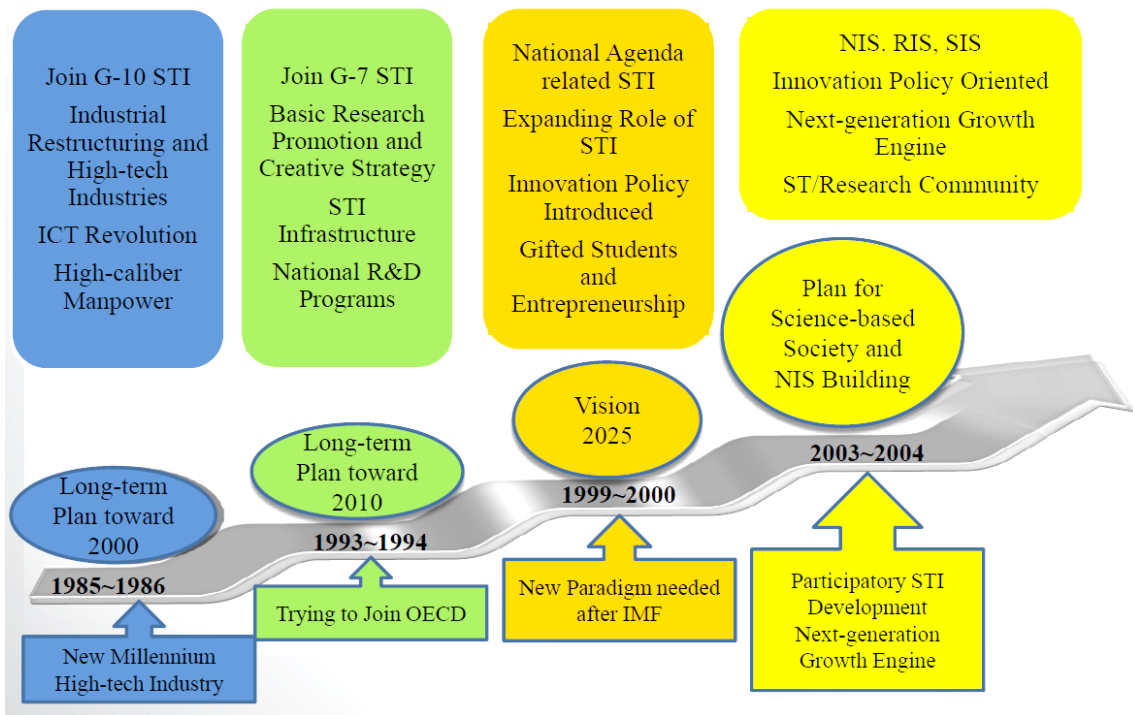
- Identification of the National Development Vision and National Development Goals (in the Overall National Plan)
- Setting of Planning Guideline and Macroeconomic and Social Development Index for the S&T Plan

- Performance of Future Foresight and S&T Forecasting
- Analysis of Current S&T Level, Capabilities, and Resources
- Finding the Gap (and Benchmarking)
- Performance of Subsectors' Development Plans
- Integration and Coordination of Subsectors' Plans
- Review and Finalization of the S&T Development Plan Draft
- Coordination and Establishment of the Plan (Cabinet Level)
- Formulation of Annual Implementation Plan
- Implementation
- Annual Performance Review/Evaluation
- Feedback and Formulation of Rolling Annual Plan

Examples of technology forecasting are listed below. The basic structure of trends/gap analysis conducted by the Korean government for the National STI Development Plan is shown in Figure 3-2.

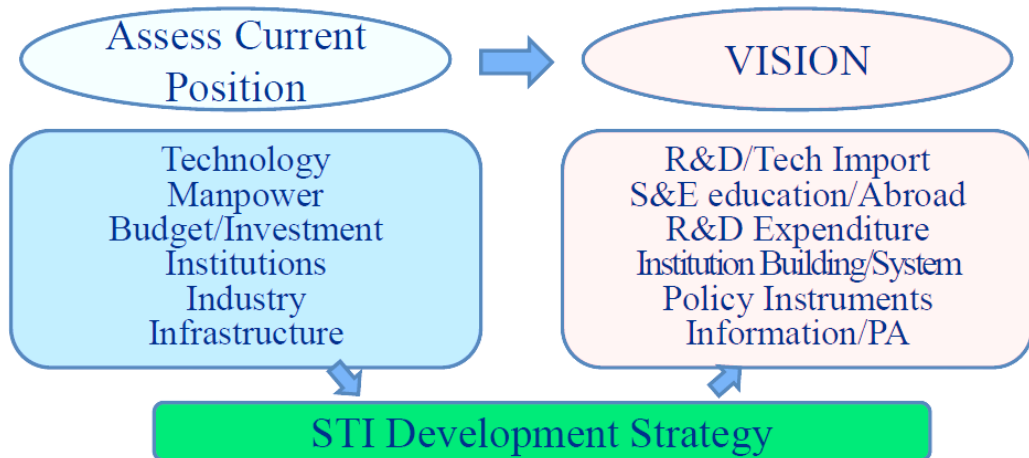
- Technology Forecasting 1995-2015 (1994)
- Technology Forecasting 2000-2025 (1999)
- Technology Forecasting 2005-2030 (2004)
- Technology Forecasting 2010-2035 (2010)
- Technology Forecasting 2015-2040 (2015)

[Figure 3-1] Overview of Four Korean Cases of Long-term STI Development Plan



Source: Young Il Park (2019)

[Figure 3-2] Korea's Gap Analysis Mechanisms



Source: Young Il Park (2019)

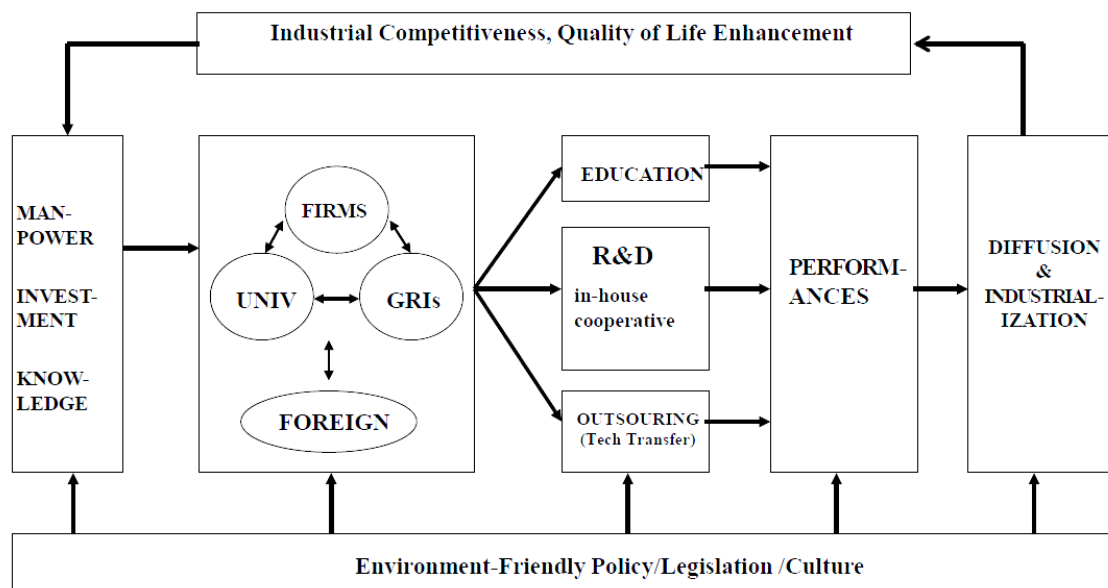
1.3.1 Key Contents of the National STI Development Plan

The basic but necessary contents that should be included in the national STI Development Plan is categorized into “input,” “throughput,” “output,” “feedback,” and “contextual.”

- Input: R&D Investment, R&D Personnel, S&T Information, National R&D Programs, etc.
- Throughput: Linkage, Policy Instruments, Legislation, Infrastructure, etc.
- Output: R&D Outcomes (3P), Industrialization, etc.
- Feedback: Planning & Coordinating Mechanism, etc.
- Contextual: Other Systems, Cultural Factors, etc.

The contents can be restructuralized again from the lens of the National Innovation System (NIS).

[Figure 3-3] Factors and Structure of the National STI Development Plan



Source: Young Il Park (2019)

Note, however, that the contents above are basic elements drawn from Korean experiences. Still, the specific contents and structural mechanism can vary according to each country's conditions.

1.3.2 Issues, Opportunities, and Challenges

When developing the national STI Development Plan, the following issues need to be considered:

- What are the expected R&R (Roles & Responsibilities) of science and technology for the future of our nation?
- What are the short-term, mid-term, and long-term visions of our nation's S&T development?
- What is the essence or nature of S&T advancement?
- Who will be leading and driving the advancement of S&T?
- How can we bring the advancement of S&T closer to reality?
- How do we utilize and deal with the internal and external environments?
- What is the best-expected ecosystem for the advancement of S&T?
- Is the current national structure appropriate for the advancement of S&T?
- What are the specific action plans for the advancement of S&T?
- What are the best ways of communication between the citizens and S&T?

When developing a strategic plan of national R&D programs, there might be some opportunities and challenges as follows (drawn from the Korean experiences):

- Enhancing the Level of National Stock of Technological and Scientific Knowledge
- Providing Driving Forces to Meet the National Agenda
- Leading Industrial Sectors – Suggesting the Future Investment Direction
- Cultivating the Necessary Human Resources & Promoting International Cooperation
- Bridging Activities Between S&T Fields and Other Sectors

1.3.3 Policy Implications

From the overall progress of the Korean experiences of establishing the national STI plan, some policy lessons below can be drawn:

- Considering the Nation's Development Stages
- Selectivity and Concentration Policy
- Innovation Policy – Dissemination, Industrialization, Entrepreneurship
- Competence of Technocrats, S&T Policymakers, and Professionals
- Importance in Institution Building
- Consensus on and Public Acceptance of S&T Development
- Cooperation and Support from Ministries

For Vietnam's future work on establishing a national STI plan, the following policy implications suggest the necessary actions for successful practice:

- Importance of Demand Survey, Technology Forecasting, and Technology Roadmap
- Enhancing National Intelligence Capabilities
- A Mechanism for Coordination and Cooperation
- Legal Framework and Institution Building
- Periodically Rolling Plans
- Participatory Planning Process

2. Technology Foresight in Korea

2.1 Introduction to Technology Foresight

2.1.1 Rationale of Technology Foresight¹⁴

Historically, mostly in developed countries when they were in the industrial development stage, technology foresight has been the critical tool for STI policymaking. In the 1970s, Japan institutionalized TF as a system for the analysis of opportunities given by new STI development to aid in long-term STI policymaking (mostly Delphi Survey). In the 1980s and especially the 1990s, a similar ambitious TF program was developed in Canada and several European countries (in the face of globalization and rising competition). International organizations -- notably the EU, APEC (particularly influential in Southeast Asia), and UNIDO (especially in Latin America and Eastern Europe) -- organized training programs, held conferences, and produced guides to best practices. Noteworthy is the fact that the EU required new members from Central and Eastern Europe to adopt and utilize Technology Foresight in developing their own practices. In some countries, it was practiced as half-hearted exercise, or “me too” fashion.

Note, however, that it is booming again due to increasing uncertainty and complexity. Recently, there have been some emerging challenges in STI policies.

- Digital Technology Evolution (capitalism without capital)
- Difficulties in assimilating new generations of technology (generating & exploiting new knowledge)
- Reconciling pressures on public finances with requests to fund new initiatives in STI
- Cumbersome and ineffectual structures of the existing governance of STI
- Social unease about some aspects of science and some applications of technology (notably biosciences and nuclear technology)

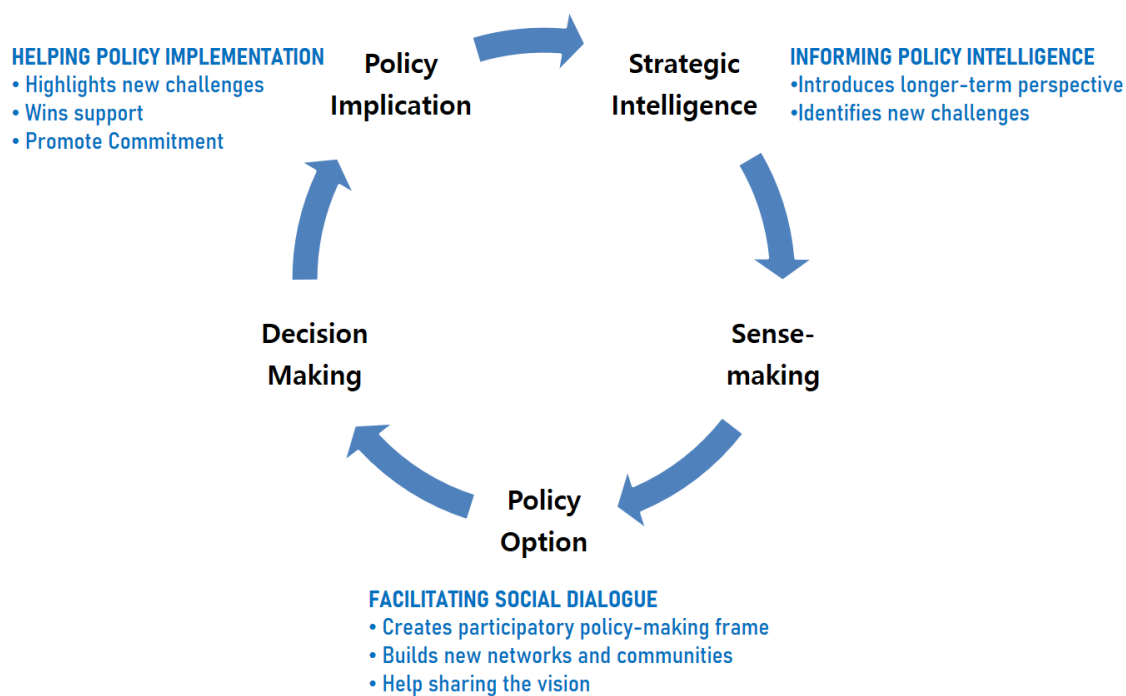
¹⁴ Byeongwon Park (2019); Hyun Yim (2019); Choi M., Choi H. (2015)

- Demands for accountability for public expenditure

In responding to these challenges, the government often faces the dilemma of decision making. Information gap occurs between the research, academe, and industry areas and government decision-making level. While the former has too much information, the latter experiences limited or abridged information. This problem is attributable to the process of simplifying the information so that distortion of data or information occurs.

In this regard, the foresight practice can enable effective, efficient policymaking. The figure below shows how foresight can help each stage of policymaking be conducted better.

[Figure 3-4] Benefits of Technology Foresight on Policymaking



Source: Byeongwon Park (2019)

2.1.2 Characteristics of (Technology) Foresight¹⁵

The rationale of technology foresight as explained in 2.1.1 comes from the characteristics of foresight.

- Acknowledgement of diverse possible futures
 - The future is not predetermined but can develop diversely according to the present efforts.
 - The future is something that is uncertain, opening to be better.
- Action-oriented
 - Foresight necessitates not only analyzing or thinking about future development but also creating it.
 - Thus, Foresight could be useful only when the results are translated into action.
- Participatory
 - Foresight is conducted not by minor professionals but diverse stakeholders.
 - The results of the TF are open to the public, which would then give feedback.
- Multi-disciplinary
 - The target problems that are expected to be solved by foresight exercises are actually not bound to one area or discipline.
 - All the possibly related variables that may have indirect or direct impacts on the issue should be considered through the multi-disciplinary approach.

These characteristics not only make foresight useful for policymaking but also suggest how to frame the foresight tasks well. In other words, these characteristics are important since they are often recalled for the basic principles and guidelines of technology foresight.

¹⁵ Yim Hyun (2019)

2.2 History of Korea's Technology Foresight¹⁶

According to Korea's Science and Technology Basic Law, Korea conducts technology foresight every five years. Implemented by the Korea Institute of S&T Evaluation and Planning (KISTEP), the project runs for one and a half years. The Korean technology foresight aims to provide the vision and direction of emerging S&T areas by identifying new technologies that may have high potential in terms of growth of national wealth and betterment of quality of human life.

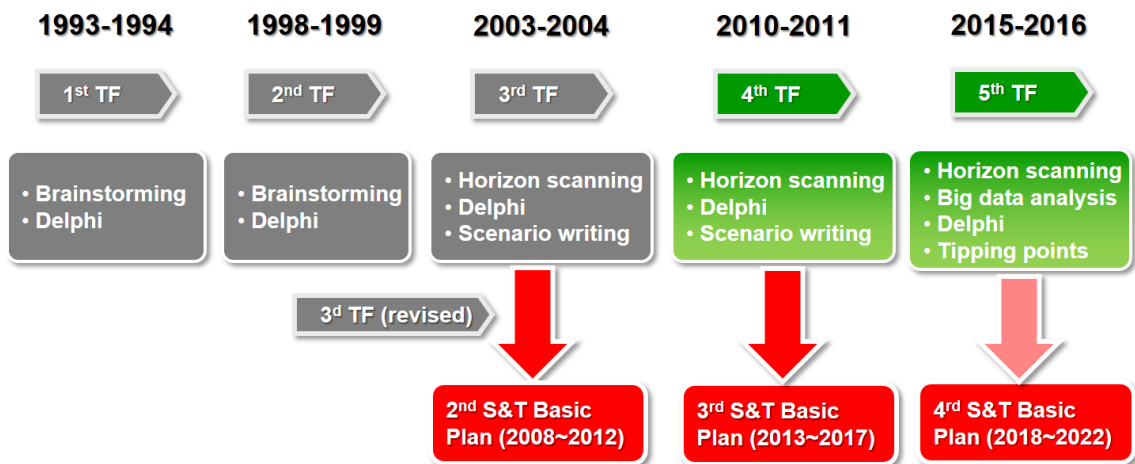
Based on the technology foresight, Korea develops a new S&T Basic Plan every five years alongside the launch of a new administration. All Science and Technology planning activities at the national level are linked to the National Science and Technology Basic Plan. The National Mid- and Long-term Science and Technology Development Strategy selects the national core strategic technologies based on the TF results on the future technologies. Based on this, the government establishes a strategic technology roadmap. The technological development of the focused initiatives of the S&T Basic Plan reflects it. The 652 future technologies identified by the fourth TF went through a review process by committees responsible for the national R&D budget as well as by R&D-related ministries. As a result, a list of 120 national strategic technologies was compiled; these technologies were also identified by the third S&T Basic Plan. The importance attributed to Korea's technology foresight (TFs) has grown with every round of TF (Choi M., Choi H., 2015).

Since the first TF for the period 1993–1994, Korean TF has been further developed and advanced in response to the socioeconomic changes and to the increasing demands for TF exercises. Since the Framework Act on Science and Technology in 2001, which specified regularly carrying out TFs, national TFs have been conducted every five years. Thus, from the third TF onward, Korea has been carrying out TFs every five years. In 2007, the third TF was revised to increase complementarities with the S&T Basic Plan, the nation's top-level plan in the field of S&T. The results of the revised TF were directly reflected to the second S&T Basic Plan. Furthermore, the results of the fourth TF (conducted during the period 2010–2011) found expression in the third S&T Basic Plan. All four TFs performed to date

¹⁶ Choi M., Choi H. (2015)

have primarily used the Delphi method. Beginning with the third TF, future social trends were first identified, and then future technologies were predicted based on these trends; moreover, scenarios were developed based on the results of the TF. Currently, the Ministry of Science, ICT, and Future Planning (MSIP) is responsible for TFs, and KISTEP conducts the TFs (Choi M., Choi H., 2015).

[Figure 3-5] History of Korea's Technology Foresight



Source: Yim Hyun (2019)

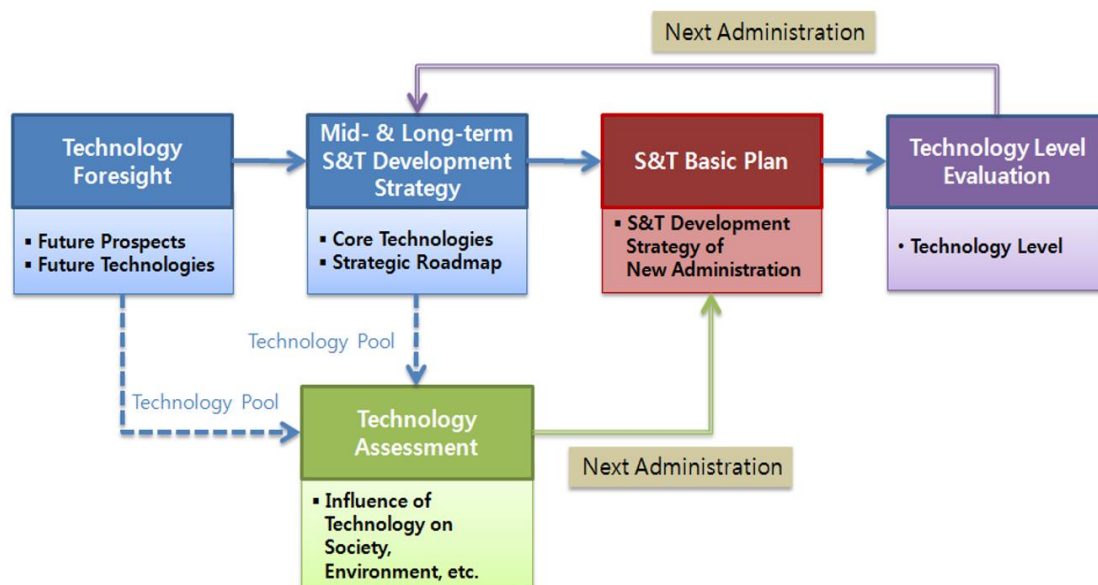
2.3 Governance and Procedures of Korea's Technology Foresight

2.3.1 Governance of Technology Foresight and National S&T Plans

As mentioned in 2.1, technology foresight is conducted for the establishment of national S&T strategies and plans. Thus, the governance of Korea's technology foresight should be understood from structuralizing the national-level S&T Planning governance.

The Korean government establishes the national S&T Basic Plan every 5 years. The national S&T plans are set based on the Basic Plan. Figure 3-6 shows the overall process of national S&T planning.

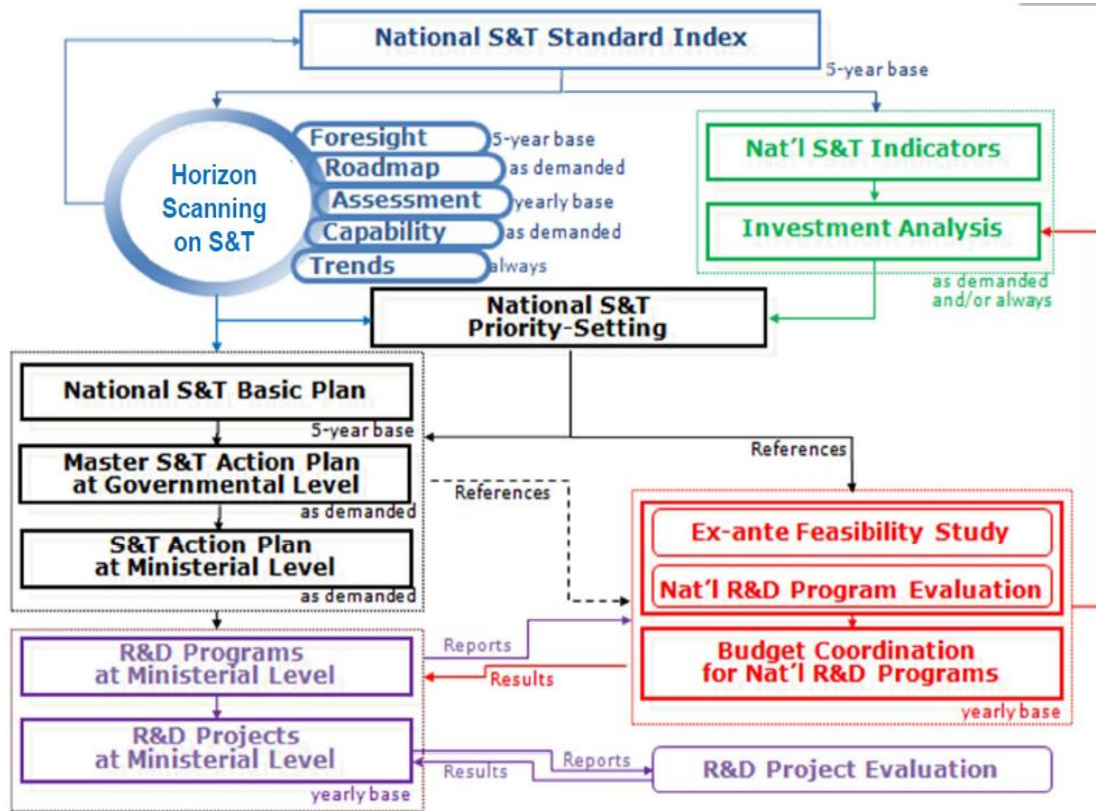
[Figure 3-6] Korean National S&T Strategy Planning



Source: Hyun Yim (2019)

Within the national-level process of S&T planning, technology foresight is implemented in each stage of planning. Figure 3-7 shows a more detailed process of implementing technology foresight in policymaking.

[Figure 3-7] Korean S&T Policymaking Process



Source: Hyun Yim (2011)

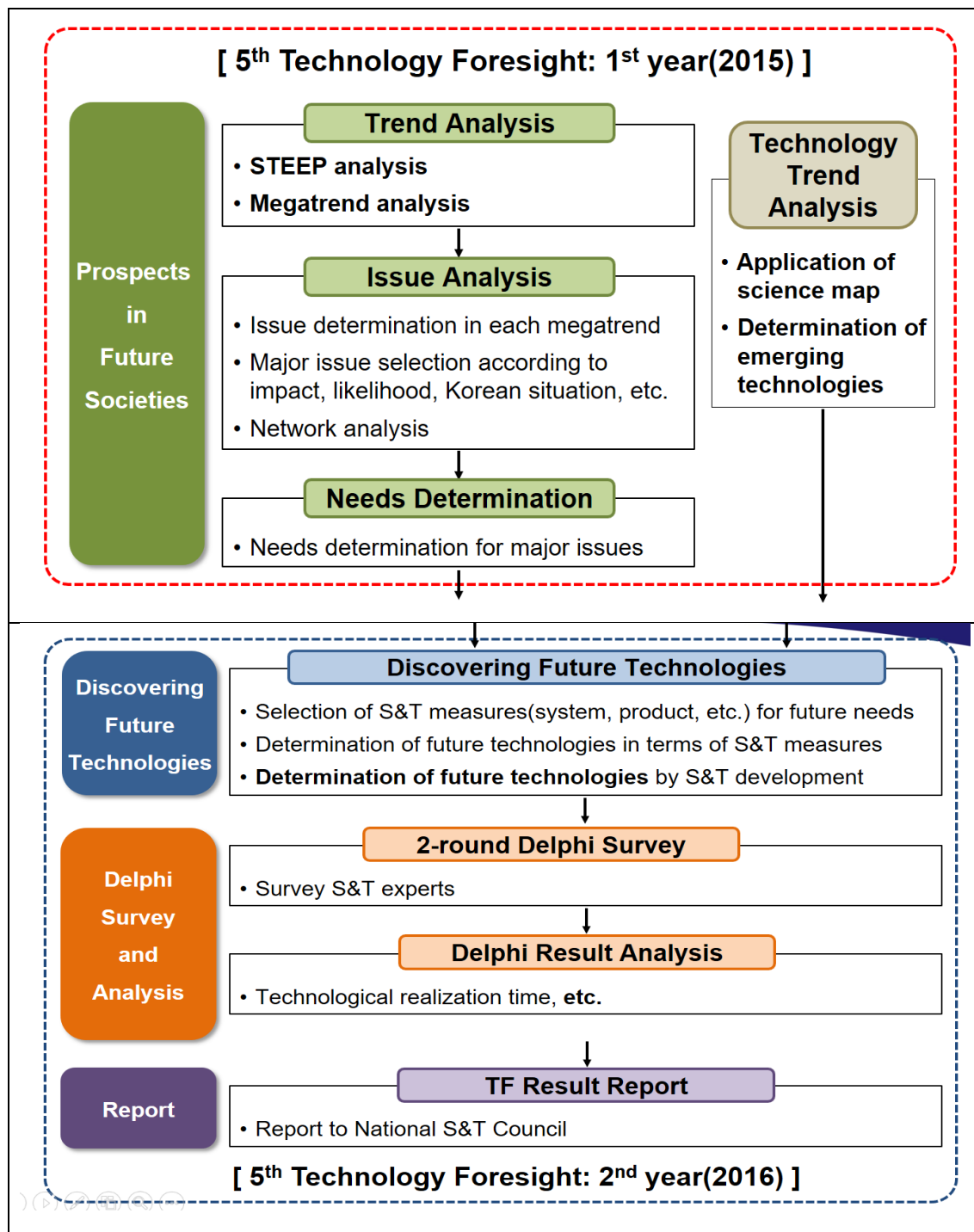
2.3.2 Procedures for Korea's Technology Foresight

In this section, the procedures for Korea's technology foresight are described with the 5th technology foresight practice as the most recent case conducted in 2015 and 2016.

The 5th Korean National Technology Foresight had four major steps: i) Prospects in future societies; ii) Discovering future technologies; iii) Delphi Survey and Analysis; and iv) Report. These overall process have been conducted for two years, one i) in 2015 and the other ii), iii), and iv) in 2016.

Figure 3-8 shows the overall process of the 5th technology foresight, with the specific procedures and results of the analysis described below.

[Figure 3-8] Process of the 5th Technology Foresight



Source: Hyun Yim (2019)

From the trend analysis, 5 megatrends and 40 trends were identified through literature review and STEEP analysis. The items of megatrends and trends are listed in the table below.

[Table 3-1] 5 Megatrends and 40 Trends

Megatrends	Trends
Human Empowerment	Increase of life expectancy, Low birth rate, Artificial intelligence and automation, Women empowerment, etc.
Innovation by Hyper-connectivity	Digital network society, Digital democracy, Hyper-connectivity technology, etc.
Anger of the Earth	Natural disaster, Food crisis, Water stress, Destruction of ecosystem, Imbalance of energy supply and demand, etc.
Evolution of Social Complexity	International conflict, Cultural diversity, Social disaster, Unification issues, International power, etc.
Reorganization of Economic System	Urbanization, Connection of global economy, New manufacturing paradigm, Low-growth economy, Chinese power, etc.

Source: Hyun Yim (2019)

In the stage of issue analysis, three sub-analyses were conducted; Issue determination, major issue selection, and network analysis.

For the issue determination, 100 future issues of each trend were determined in terms of time period, i.e., within 10 years and from 11 to 25 years. Those issues were then divided into 2 categories: 65 short-term issues and 35 long-term ones.

The major issue selection involved a general public survey. The purposes of this step are 1) to determine major issues that greatly impact Korean society and 2) to identify the needs and future technologies.

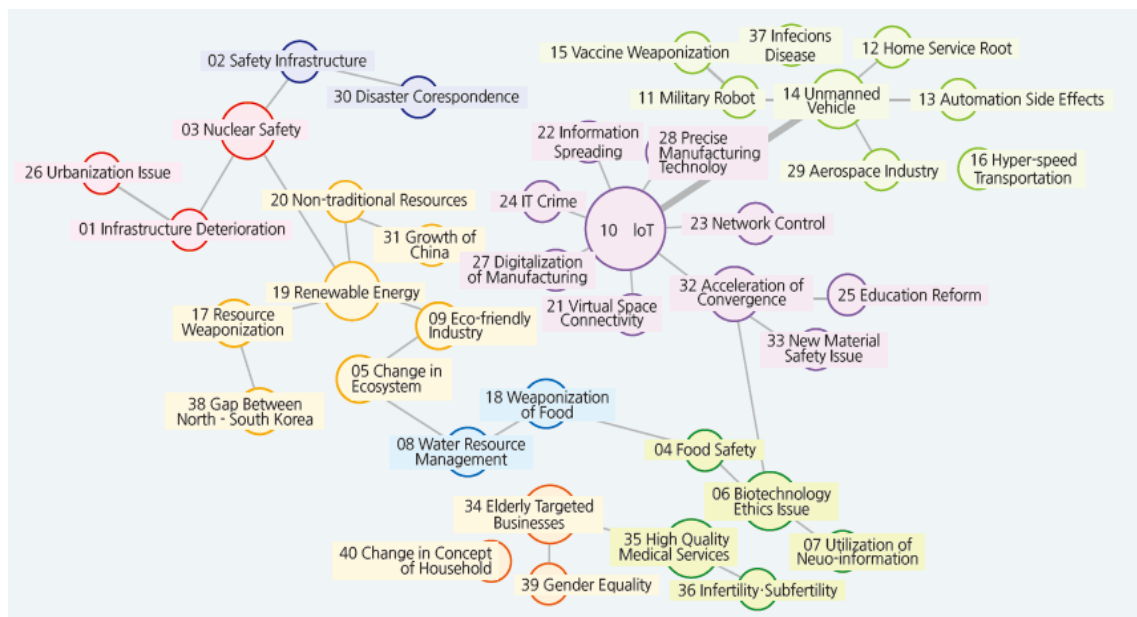
The general public survey was conducted via email. The survey was sent to over 10,000 members of major S&T portals, and 473 respondents responded to select the major issues of each megatrend. The selection criteria were "likelihood," "impact," and "possibility of

S&T measures.” For the next step, 100 issues selected in the issue determination were evaluated by the foresight committee considering the survey results. As a result, 40 issues were selected as major issues.

For the 40 major issues, network analysis was conducted with text mining methodology based on natural language processing. The data included 26,108,658 news articles collected from portals and press homepage (newspaper, broadcasting news, Internet news, etc.). The detailed process involved the collection of news articles, issue keywords selection, natural language processing, and network and associative term analysis in that order. The analysis resulted in the identification of the most referred to issue, such as crime by information technology, green industry, renewable energy, IoT, infectious disease, and nuclear safety. Issues interest in which increased the most from 2012-2013 to 2014-2015 were IoT, cutting-edge manufacturing, unmanned vehicle, manufacturing paradigm shift, safety infrastructure, disaster risk reduction, and infectious disease.

For the next step, **6 Major Issue Groups** were derived based on their correlation according to Big Data-based network analysis as shown in the figure below.

[Figure 3-9] Deriving Key Connections in the Issue Network (2012-2015)



Source: Hyun Yim (2019)

At the “needs analysis” stage, needs specific to major issues were derived. Based on an in-depth analysis of each major issue, the detailed socioeconomic demands of the future society in terms of scientific and technological responses were identified.

Based on the analysis, the needs from each major issue were drawn through the departmental meetings of the foresight committee. In the 40 major issues, there are 231 needs. The needs were drawn with average of 5.8 from each issue.

Technology trend analysis involves two steps: one is the application of science map, and the other is the determination of emerging technologies. The technology trend analysis aims to 1) identify connection and development trends among research areas and 2) investigate emerging research areas and technologies by time-series analysis and apply the results to determine technology push-type future technologies. Two methods were applied. **One is the network** analysis of terms in the title and abstract of publications. The other is time-series analysis from 2000 to 2014 (3 stages of 5-year interval). The subject of analysis was 39 Scopus journals of 13 research fields.

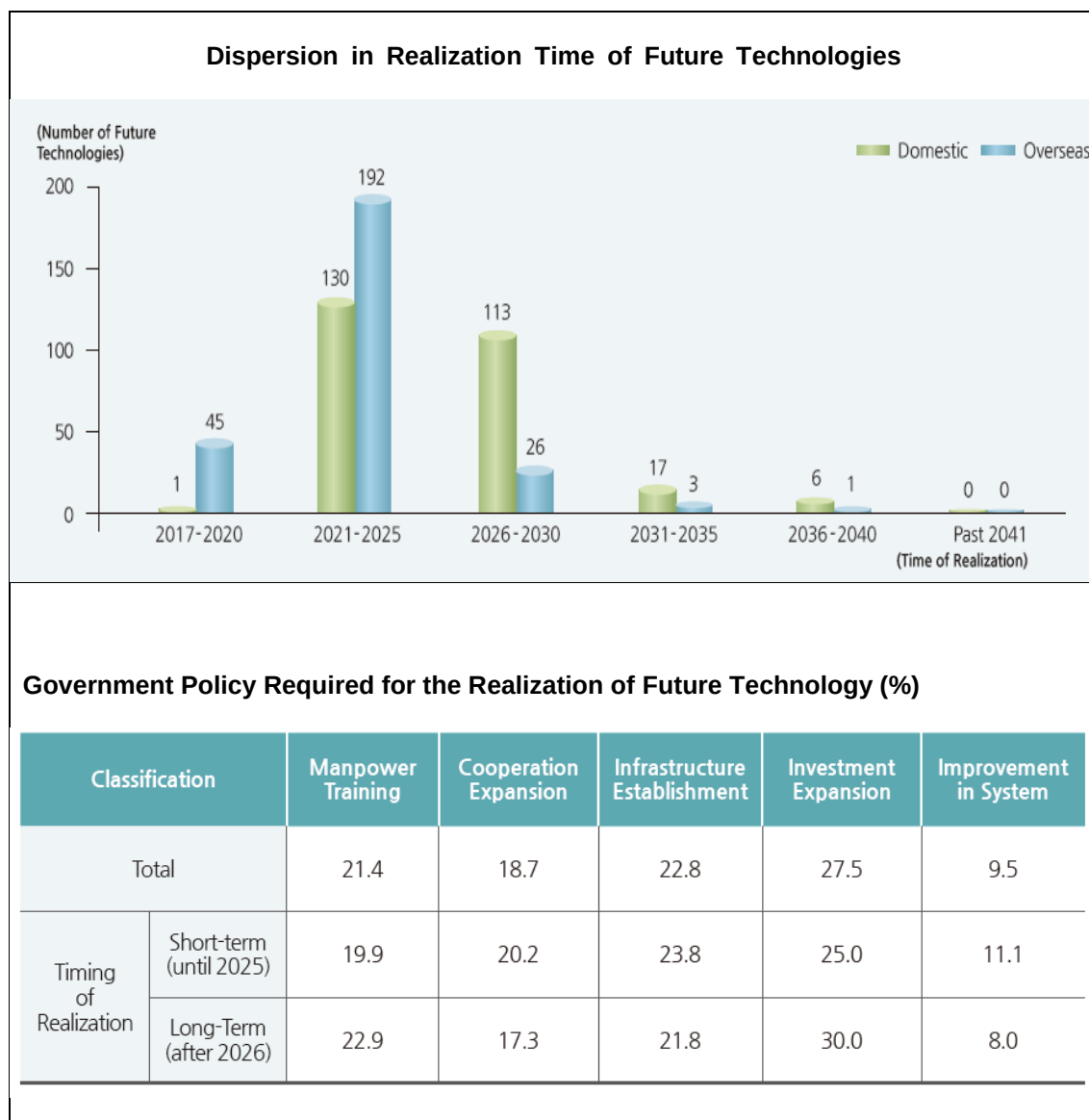
From the analysis, a total of 267 future technologies were derived. The candidates were selected by reviewing the results of analysis and interpretation of a needs and knowledge map. The final future technologies were obtained by revising and adjusting them.

In 2016 as the second year, a Delphi survey was conducted. The overview of the survey is as follows:

- 2-round Delphi Survey
 - 4,133 respondents (12,884 responses) participated in the 1st survey.
 - 3,621 respondents (11,085 responses) participated in the second survey, for a follow-up rate of 87.6%
- 267 future technologies were examined to assess their “Realization time,” “Required government policies,” “Innovativeness and uncertainty of future technology,” and “significance and technical competitiveness of future technology.”

The results of the survey on each assessment item are shown below.

[Figure 3-10] Results of the Delphi Survey



Technologies that are High in Both Innovativeness and Uncertainty (Group I, Top 10 Techs, 5-point scale)

Tech No.	Future Technology	Innovativeness	Uncertainty
10	Local Climatic Damage Mitigation Technology Minimizing Surrounding Environment Effects	4.2	4.4
128	Manned Lunar and Mars Base Construction Technology	4.6	4.3
129	Asteroid and Lunar Space Resources Mining Technology	4.5	4.3
216	DNA Chip-based Mass Data Storage and Information Management Technology	4.5	4.2
127	Crop Development Technology for Outer Space Farming	4.4	4.1
145	Memory Scanning, Storage and Modification Technology	4.4	4.1
246	Quantum Computing for Superspeed Calculation	4.7	4.1
124	Self-Propellable, Self-Operable Spacecraft	4.6	4.0
92	Outer-space Photovoltaic Technology	4.7	4.0
153	Highly Accurate Neural Signal-Based Personal Authentication Technology for Neuro-information Security	4.5	4.0

Technologies that are High in Both Significance and Competitiveness (Group I, Top 10 Techs, 5-point scale)

Tech No.	Future Technology	Technical Competitiveness	Significance
133	Intelligent Exploration Robots Capable of Overcoming Obstacles	3.7	4.2
79	Greenhouse Gas Free Hydrogen Reduction Steel Making Technology	3.6	4.3
134	Self-combining and decombining Robot for Optimized Performance in Adverse Conditions	3.6	4.3
159	Ultra-Sensitive Microfluidics Biosensor System for Multi-Blood Analysis	3.5	4.3
189	Ultra-fine 3D High-functioning Element Manufactured through 3D Printing	3.5	4.3
112	Highly Reliable and Safe Integrated Manned/Unmanned Aircraft Monitoring System	3.5	4.4
260	Advanced Quantum Cryptosystem Key Distribution Technology Preventing Inverse Calculation	3.5	4.5
172	Next-Generation Gene Analysis (NGS) Technology	3.5	4.2
193	3D Printing Specialized Designing Technology	3.5	4.3
212	Alternative Technology to Radioisotope Testing in Non-Destructive Testing	3.4	4.4

Source: Hyun Yim (2019)

2.4 Policy Implications

From the Korean case review, some major policy implications can be drawn. There are some common features of technology foresight exercises that can be found in many other countries' cases aside from Korea's case. At the same time, there are some policy lessons that can be relatively more suitable for Vietnam. This section suggests the main implications from those two aspects.

First of all, most of the Technology Foresight (TF) exercises have similar or common features.

- TF is designed to wield an influence on STI policy. Therefore, having access to the decision makers or high-level officials who may influence on the decision making of STI policy is essential. In many cases, the decision makers are the sponsors of the exercises. The expected influence can be identifying priority areas for R & D or training.
- TF plays a role in addressing weaknesses in the innovation system, especially in the process of aligning different stakeholders around shared appraisals of future prospects.
- Typical products will include: codified reports, presentations, Internet resources, etc.
- TF involves “process benefits” by connecting the dots among stakeholders and, equally important, enabling key actors.
- TF is liable to be participatory and to establish networks among various sources of knowledge of different regions and institutions.
- TF is prospective by definition as it takes a longer-term view than the usual policymaking does.

There are some other specific lessons for Vietnam and other countries that try to adopt or develop their technology foresight capacity and policies:

- There is no one-size-fits-all methodology.
 - Regard it as an essential point of TF exercises by considering the local conditions and circumstances.

- Implementing technology foresight to policy actions will be very contingent on circumstances (while easy to identify the decisions that need to be taken).
- There might be more benefits gained from being more informed by foreign experiences due to the global convergence trends over the past few decades (3As: adopt, adapt, advance).
- When the national innovation system has more functional components in it, the effectiveness of TF projects would increase.
- Having consultation from a wider range of experiences, involving many aspects of domain beyond one specialized features, which would also engage broader sectors and stakeholders.

3.

Korean STI Institutional Strategies –KIST case

3.1 Rationale of Technology Planning for Institutional Strategy¹⁷

3.1.1 Background

In Korea, Government Research Institutes have several common missions:

- To secure competitiveness
- To acquire national talents
- To prepare for the future
- To nurture on-site talents
- To develop open research platforms

As the leading S&T player in Korea, GRIs have to respond to the dynamics of innovation and technology changes. At the same time, since the funding comes from the government (fundamentally from TAXES), GRIs have the mission of serving society and country based on the national strategies. Thus, in response to the recent technological disruption and national S&T paradigm, GRIs have two parallel missions of frontier R&D and problem-solving R&D. The former is more focused on basic-original, and the latter is for the practical solution.

3.1.2 KIST's institutional strategies

As a leading research institute in Korea, KIST has its distinguished vision and missions that reflect the dynamics of science, technology, and innovation. The institutional vision and strategies are as follows:

- Vision: Beyond the MIRACLE with PRIDE

¹⁷ Yoohyung Won (2019); KIST website

- Beyond the MIRACLE: "Beyond the glory of the last half-century, let us create a new miracle for the next half-century." MIRACLE stands for the seven areas that KIST should focus on (Material, Information, Robotics, Agriculture, Carbon cycle, Life science, Environment).
- Core Values of Vision
 - Research Excellence: Develop future leading source technology and make operations conform to global standards.
 - Open Innovation: Build an open platform for convergence/collaboration and create infrastructure for communication/sharing.
 - Public Purpose: Develop core technology in line with the national agenda and promote an office culture of ethics and integrity.
- Strategies
 - P: Pioneer – Develop a world-class operating system and promote challenge in research.
 - R: Revolution – Lead the 4th Industrial Revolution and provide solutions for S&T challenges.
 - I: Internationalization – Enhance Korea's S&T status and share expertise with developing nations.
 - D: Duty – Create an office culture of integrity and promote S&T awareness
 - E: Entrepreneurship – Support innovative startups/SMEs and create innovation clusters

The key strategic questions of technology planning for KIST are:

- What is the core and emerging technology globally?
- Who leads this technology?
- Where are we positioned?

Based on the questions above, KIST selected its strategic foci:

- Problem solving: Urgency, Solid needs, Internal competency
- Frontier research: Prominent impact, Radical novelty, Relatively fast growth

3.2 R&D Planning of KIST¹⁸

3.2.1 Planning Strategies

KIST is the globally prominent think tank that is ranked 6th among GRIs in the world, according to Thomson Reuter in 2017.

KIST has strengthened its capacity by transforming its missions and roles in the national innovation system.

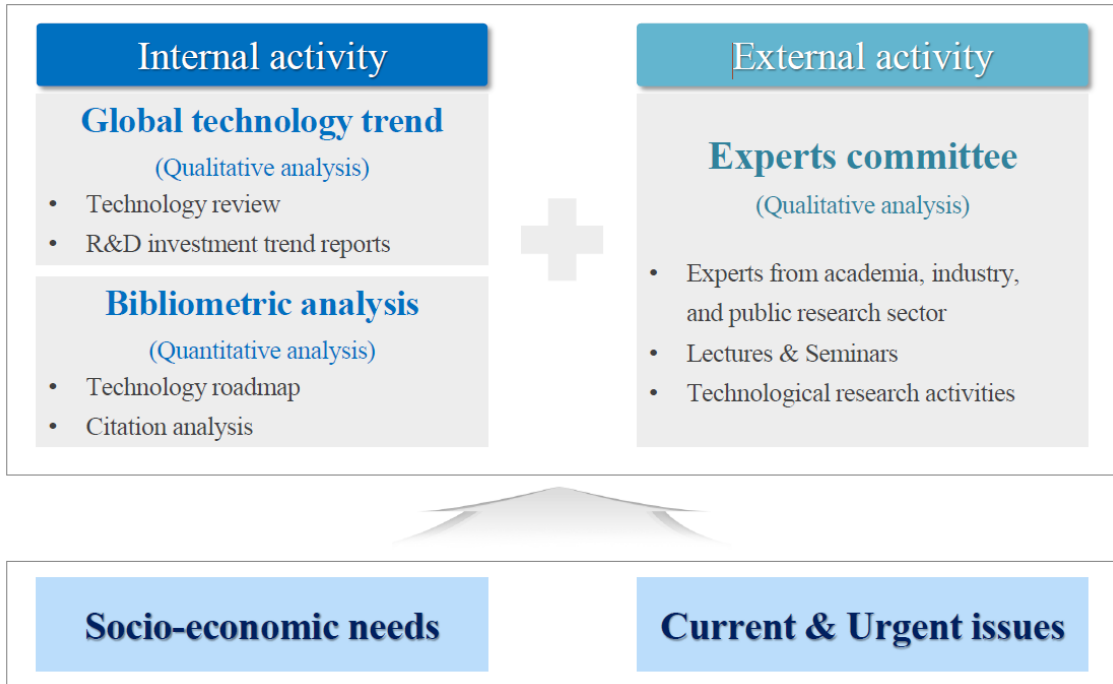
During the 1960s ~ 1970s, its mission was to support major industries such as manufacturing, heavy chemical industry, etc. During the 1980s and 1990s, it expanded its research area to advanced technologies such as artificial diamond, doping control and lyocell fiber, etc. Since the 2000s, it has further expanded and advanced its research scope to the fundamental technology area, such as digestive endoscope, brain science, robot technologies, etc.

Like the foregoing, KIST has responded to the environmental changes and identified its roles as required by the country and society.

The key success factor is its planning strategies and governance.

¹⁸ Yoohyung Won (2019); KIST website

[Figure 3-11] KIST's Technology Planning Activities



Source: Yoohyung Won (2019)

3.2.2 Strategic programs

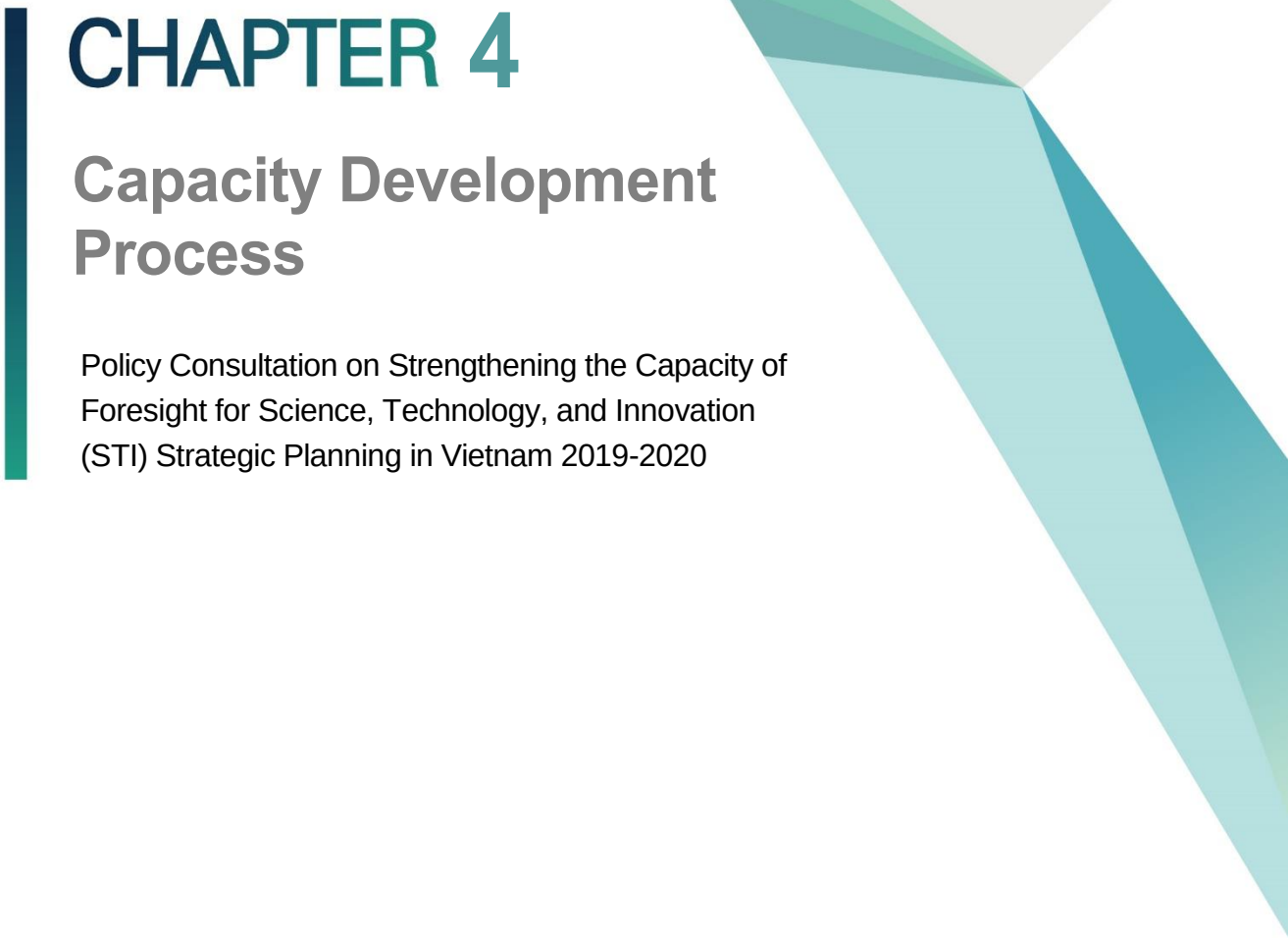
Based on the strategic planning capacity, KIST has enhanced its research areas. The programs were designed according to the missions and strategic foci. The major programs for each strategic focus are as follows:

For Problem Solving,

- Alzheimer's Disease Diagnosis
 - It aims to respond to the social needs arising from the aging society and increasing social cost.
 - KIST pursues convergence research with neuroscience, brain medicine, and neuropathy.
 - KIST has Big Data-based Alzheimer's disease coordination, platform implementation, and life school robot development.

For Frontier Technologies,

- Quantum computing
 - It aims to develop technologies beyond the recent trends.
 - The recent trends are focused on **Computing-based Technologies** such as AI, Block Chain, and Big Data.
 - As a preemptive response, KIST researches on Quantum computing to solve **security problems** caused by the era of intelligence.

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing left, and several overlapping triangles in shades of teal and light blue pointing right.

CHAPTER 4

Capacity Development Process

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation
(STI) Strategic Planning in Vietnam 2019-2020

Chapter 4. Capacity Development Process

1. Kick-off Meeting and Field Visit

1.1 Introduction (objectives, team, schedule, etc.)

1.1.1 Objectives

The purpose of the first visit is to have an inception meeting and a workshop for knowledge sharing. The inception meeting shall involve project counterparts, VISTI, and STEPI to discuss the project purpose, activities, and expected outputs. The workshop, titled “National Future Strategy and Foresight,” will be a public event for sharing knowledge on the national STI strategic foresight plans or activities of Vietnam and Korea. Various stakeholders including government officials from the relevant ministries and the directors of the research institutions will be invited to the event to build consensus on the importance of strengthening the national strategic planning system and the directions for foresight capacity building. This event will discover policy demands for the main institutions and ministries.

The following are the three main goals of the visit:

- Confirm the project purpose and expected outcome
- Collect the necessary information for assessing Vietnam’s national STI strategic planning and technology foresight capacities
- Share Korean experiences in national strategic planning and technology foresight
- Exchange ideas and opinions on capacity-building programs

The main activities of the workshop will be as follows:

- Short remarks from the STEPI project leader on the project overview including the purposes, scope, methodology, implementation, planned schedule, and results
- Presentations by the Korean experts on Korean cases of national strategic planning and technology foresight
- Presentations by the Vietnamese experts on Vietnam's STI policy and strategy, the development and application of technology foresight, and a few cases, to be followed by a lively discussion on key issues and the exchange of ideas and opinions
- Follow-up in-depth meetings with stakeholders to collect their personal views and insights on improving foresight and strategic planning capacities.

1.1.2 Team

The research team was responsible for holding the kick-off meeting and field visit to Vietnam. The research team conducted interviews with the stakeholders from VISTI and other research institutions and decision makers including the Deputy Minister of Science and Technology of Vietnam.

[Table 4-1] Kick-off Meeting and Field Visit Research Team

Organization	Name	Project role/Position	Contact
STEPI	Dr. Jungwon Lee	Project Leader Senior Research Fellow	leejw@stepi.re.kr
	Dr. Byeongwon Park	Foresight Expert Research Fellow	bwonpark@stepi.re.kr
	Ms. Yongin Choi	Project Coordinator Researcher	luciachoi@stepi.re.kr
Ewha University	Prof. Young Il Park	Adviser / Speaker Professor	yipark@ewha.ac.kr
KIST	Dr. Yoohyung Won	Adviser / Speaker Principal Researcher	yhwon@kist.re.kr

1.1.3 Schedule

[Table 4-2] Kick-off Meeting and Field Visit Schedule

Date/Time		Institution	Description
Start	End		Activities and Time
DAY 1: Monday, April 15, 2019			
		Arrival	ICN-HAN (Jungwon Lee, Yong-in Choi)) 21:50 by KE0679 FRA-HAN (Byeongwon Park) 05:45+1 by VN0036
DAY 2: Tuesday, April 16, 2019			
09:00	12:00	Hotel	Internal Meeting – STEPI team
12:00	14:00		Lunch
14:00	17:00	Korean Embassy to Vietnam	Interview with the Ambassador and Counselor - Introduction of the project - Discuss Vietnam’s STI policy issues and expected outcomes - Discuss the <u>agenda</u> for further cooperation on the project
DAY 3: Wednesday, April 17, 2019			
09:30	12:00	KOICA	Interview with the Representative - Introduction of the project - Discuss Vietnam’s STI policy issues and relevant ODA projects - Discuss the <u>agenda</u> for further cooperation on the project
12:00	14:00		Lunch
14:00	17:00	<u>VISTI</u>	Inception Meeting - Introduction of the project - Discuss the project purpose, <u>main</u> activities, schedule, and expected outcome - Exchange ideas and views on the project details - Identification of policy demands for the project
		Arrival	ICN-HAN (Yoohyung Won) 13:05 by KE0483 ICH-HAN (Young Il Park) 21:50 by KE0679

Date/Time		Institution	Description
Start	End		Activities and Time
DAY 4: Thursday, April 18, 2019			
09:30	12:30	Army GuestHouse Hotel	Workshop on National Future Strategy and Technology Foresight
12:30	14:00		Lunch
14:00	17:00	VISTI	In-depth Discussion - Discuss the results of the workshop - Interview with various internal focused groups - Identify key challenges in policy and in the project - Exchange views and insights on the current status and capacity-building programs for Technology Foresight and Strategic Planning
DAY 5: Friday, April 19, 2019			
11:00	16:00	VKIST	Interview with the Representative - Introduction of the project - Discuss Vietnam’s STI policy issues and relevant ODA projects - Discuss the agenda for further cooperation on the project
		Departure	HAN-ICN (Jungwon Lee, Yong-in Choi, Young Il Park, Yoohyung Won) 23:10 by KE0680 HAN-FRA (Byeongwon Park) 22:55 by VN0037

1.2 Kick-off Meeting

1.2.1 Kick-off Meeting and Field Visit Participants' List

[Table 4-3] Kick-off Meeting and Field Visit Participants' List

Name	Organization
Kim Jin Oh	KOICA Vietnam Office / Chief Officer
Lim In Young	KOICA Vietnam Office / Deputy Chief Officer
Kim Dong Bae	Korean Embassy to Vietnam / Counselor
Shin Dong Chan	Korean Embassy to Vietnam / Deputy Director
Shin Sang Yeol	Korean Embassy to Vietnam / Director
Ma. Nguyen Hoang Hai	VISTI / Director
Bach Tan Sinh	VISTI / Director
Kum Dong Hwa	VKIST / President
Hoang Van Cuong	Central Institute for Economic Management
To Mai Trinh	Department of International Cooperation
Yang Hi Sang	Hanoi University of S&T
Phan Van Nghia	Ministry of Information and Communication
Le Dinh Tien	Ministry of Science and Technology
Luu Quang Minh	Ministry of Science and Technology
Giang Manh Khoi	National Center for Technological Progress
Tran Hong Minh	National Center for Technological Progress
Nguyen Viet Hoa	National Institute of S&T Policy and Strategy Studies
Ta Doan Trinh	National Institute of S&T Policy and Strategy Studies
Duong Khanh Dung	National Institute of Information and Communications Strategy
Hoang Anh	National Office of Intellectual Property of Vietnam
Tran Khac Trong Tai	National Office of Intellectual Property of Vietnam
Pham Quang Tri	National Technology Innovation Fund
Ngo Ha	Science and Development Newspaper
Le Trong Tai	State Agency for Technology Innovation

Name	Organization
Nguyen Duc Hoang	State Agency for Technology Innovation
Le Trong Nghi	TDST
Nguyen Hanh	Vietnam Center for Science and Technology Communication
Ngo Thi Loan	Vietnam Center for Science and Technology Evaluation
Pham Hong Bach	Vietnam Center for Science and Technology Evaluation
Doan Thi Thu Ha	Vietnam Institute for Development Strategies
Nguyen Thi Nguyet	Vietnam Institute for Development Strategies
Vu Thi Hai	Vietnam Intellectual Property Research Institute
Vu Van Hung	Vietnam Journal of Science, Technology, and Engineering
Le Ngoc My	VISTI
Nguyen Van Hoa	VISTI
Nguyen Van Khuong	VISTI
Nguyen Viet Hung	VISTI
Tran Van Tam	VISTI
Le Vu Toan	VISTI

1.2.2 Kick-off Meeting's Main Activities and Agendas

At the kick-off meeting with VISTI, the objectives and scope of the Project were discussed. Further specific action plans were discussed, and STEPI and VISTI agreed on the main items of the project programs and activities. Since the kick-off meeting was held after the workshop on national future strategy and technology foresight, the researchers of both counterparts shared insights on Korea's S&T strategic planning and its implications on Vietnam's future policy direction. The key points of the meeting can be summarized as follows:

- Agreed on the Project Action Plan
- Discussed VISTI and other institutions' role in Vietnam's STI system
- Shared Vietnam's challenges in technology foresight and national STI strategies

- Shared Korean policy experiences in national STI strategies based on technology foresight

1.3 Field Visit (information by organization to visit)

1.3.1 Meeting with the Embassy of the Republic of Korea in Hanoi, Vietnam

A meeting with the Korean Embassy in Vietnam was held on April 16, 2019 to provide an introduction of the Project. The Korean Embassy shared the current status of Vietnam's STI policies and the STI-related ODA program. The key results of the meeting are as follows:

- Current status of Vietnam's STI policy and ODA programs
 - The main challenges of Vietnam's STI policy are "unclarified direction and insufficient contents," "STI-related data system," and "insufficient R&D evaluation system."
 - The main challenges of Vietnam's ODA programs (as a recipient country) are "sustainability" and "redundancy of programs."
- Issues to be considered to maximize the effectiveness of the Project
 - Cooperation needs from Korean R&D companies located in Vietnam
 - Cooperation opportunities with V-KIST and other ODA projects of Korea for Vietnam

1.3.2 Meeting with KOICA Vietnam Office

The research team had a meeting with the KOICA Vietnam office. The basic information of the Project were shared, and additional issues on Vietnam's policies were discussed.

- Possibility of cooperation with KOICA
 - Prioritize identifying cooperation opportunities with V-KIST

- Discuss possible ways of cooperative activities considering the role of V-KIST in Vietnam's national STI system
- Recently, KOICA has not conducted STI-related ODA projects for Vietnam as there is little demand from the Vietnamese government on the STI field.
- Issues to be considered
 - The capacity of human resources of Vietnam is much more competitive compared to that of other South Asian countries
 - For the effective progress of the Project, the local experts need to be involved actively
 - Considering Vietnam's administrative customs and culture, clear agreement on the Project budget allocation and cooperative response to document-based process are needed

1.3.3 Meeting with Vietnam's Deputy Minister of Science and Technology and VKIST

The research team had a meeting with Vietnam's Deputy Minister of Science and Technology and the President of VKIST. The in-depth discussion highlighted the following key points:

- Current status of Vietnam's technology foresight and S&T capacity
 - The Vietnamese government is trying to increase the effectiveness of implementing the national S&T policies by drawing up the quantitative goals and specific action plans in National Plan 2021-2030
 - Vietnam needs to strengthen the science and technology policy capacity of Vietnamese policymakers
 - Vietnamese research institutions need to be systemized with roles that are distinct from each other

- Several ways to enhance the effectiveness of the Project
 - SATI and VISTI can assume a role in the Project, since each has data and human resources related to technology foresight
 - The Project could link its activities with other small and large national projects that involve technology foresight and innovation policies
- Expected outcome of the Project
 - The Project, which aims to enhance Vietnam's capacity for technology foresight, can be the steppingstone for further cooperation activities between Korea and Vietnam on S&T policies.
 - The Project can be linked with VKIST's planned role expansion in analyzing Vietnam's economic and technological capacities

1.4 Workshop on National Future Strategy and Technology Foresight

1.4.1 Concept note

a) Background

Technology development is much faster and broader than many people realize. It affects every aspect of human life. Digital technology revolution such as artificial intelligence, big data, Internet of Things, block chain technology, 5G, etc. even opens up a new possibility in triggering industrial transformation. It may result in shifting the global order or solidifying the current one. It is unclear at the moment who will be the next winner.

The future is shaped and influenced by the decisions made today. Still, making the right decision at the right time to exploit the full potential of new technology and innovation becomes more difficult and probably almost impossible. The time allowed for decision making is getting shorter in contrast to the increasing degree of complexity and uncertainty of issues. Though the volume of data is getting huge, insight by sense making is too shallow. The only way to adapt and survive in the future is to embrace the complexity and

uncertainty as the inherent characteristics of the system instead of trying to predict the future. It is impossible to predict the future by forecasting through trend extrapolations.

b) Foresight capacity to shape the future

Foresight refers to approaches to informing decisions made robustly by improving inputs concerning the longer-term future and by engaging the wider social stakeholders and linking strategic vision to present-day actions. Foresight will enhance the capacity for decision making and promote public debates by providing opportunities to discuss longer-term trends as well as the expected outcome or implications of short-term decisions. Likewise, foresight provides alternative interactive learning platforms for robust decision making in the context of 4th industrial revolution-associated uncertainties.

c) Goal of the project

Nonetheless, realizing the importance of long-term thinking and the need for running foresight exercise is different from the action on vision from the foresight results. In many cases, foresight has limited influence on changing the behavior and action of the decision makers due to many causes such as unclear goals, governance, incentives and cultures, budgets, etc.

Since the lack of competence among decision makers turns out to be one of the main reasons, the primary goal of the current project is set to enhance the foresight capacity of the decision makers through the pilot-scale foresight exercise, focusing mainly on research and innovation-driven future development. The opportunities provided by such innovations intimately shape most areas of human social life.

Structured approaches are employed to build capacities in foresight, engagement, and integration among the decision makers. The approaches will be interactive and participative, focusing on long-term social, economic, and technological developments relevant to the country. Based on the project's goal, this workshop is designed to be the first step of the project for sharing the current status of Vietnam's national S&T policy and Korea's experiences in strategic planning and technology foresight. All the participants are invited to discuss the agenda and its implications to Vietnam's national STI strategy.

1.4.2 Program

Date: Thursday, April 18, 2019

Time: 9:00am – 12:40pm

Venue: Army Hotel, 33 C Pham Ngu Lao, Hanoi, Vietnam

Expected Participants: STEPI, VISTI, VKIST, Government officials, other relevant experts

[Table 4-4] Program of Workshop on National Future Strategy and Technology Foresight

Time	Activity	Speakers
8:40 - 9:00	Registration	
9:00 - 9:15	Opening Session	
9:00 - 9:05	Welcome Remarks Introduction of the project	Dr. Jungwon Lee Senior Research Fellow, STEPI
9:05 - 9:10	Welcome Remarks	Dr. Dongwha Kum President of VKIST
9:10 - 9:15	Opening Remarks	Dr. Nguyen Hoang Hai , Acting Director, Department of Science Administrator and Training, VISTI
9:15 - 12:30	Presentation Session	
9:15- 9:55	Presentation 1 Introduction to National Strategic Planning and Korean Case of STI Development - Overview of 4 Korean cases of long-term STI Development Plan - Basic framework for the STI Development Plan - Key issues, opportunities, and challenges - Lessons learned and implications for the next steps	Prof. Young Il Park Professor, Ewha University Former Vice-minister, MOST
9:55- 10:25	Presentation 2 Introduction of Technology Foresight and Its Implementation - Understanding of social change - Reasons for the recent boom of foresight - Procedure for foresight - Linking foresight to policymaking - Technology foresight to wiring NIS	Dr. Byeongwon Park Research Fellow, STEPI

Time	Activity	Speakers
	* In this presentation, the success factors and barriers of the foresight exercise from the practitioner's perspective will be given to stimulate further discussion.	
10:25-10:55	Presentation 3 How Korean GRIs are Preparing for the Future - KIST's Strategy for Redefining R&R - KIST's Alignment with the National STI Strategic Goals - KIST's Contributions to Korean STI Policy : K-DARPA Program	Dr. Yoohyung Won Principal Researcher, KIST
10:55-11:10	<i>Coffee Break</i>	
11:10 - 11:30	Presentation 4 Overview of Vietnam's STI policy strategy - Some general context for STI policy and strategy process for Vietnam - Case of using Foresight for STI Strategy 2020: some experiences - Potential implications	Dr. Tran Ngoc Ca Former Vice-President of NISTPASS
11:30-11:50	Presentation 5 Development and Application of Technology Foresight in Vietnam - Overview of the current stage of developing and applying technology foresight in a few sectors - Lessons learned for making this tool more effective in the context of transforming the current strategic mid-term and long-term planning of Science and Technology toward the engagement of wider stakeholders	Dr. Bach Tan Sinh Senior Research Fellow, VISTI
11:50-12:10	Presentation 6 Vietnam's Technology Foresight Case - Vietnam's Future Digital Economy Project - Approach applied - Recent development of Vietnam during the course of the 4th Industrial Revolution and digital transformation - Some major trends and formulation of development scenarios - Some results from modeling and recommendations	Dr. Nguyen Duc Hoang Senior Officer, State Agency for Technology Innovation (SATI)/MOST
12:10 - 12:40	Floor Discussion	All Participants
12:40 – 12:45	Closing Remarks	
	Lunch	Army Hotel

1.5 Follow-up activity (Meeting Recap)

The follow-up document was developed in April 2019 and signed by STEPI and VISTI. The key actions required for the next steps are included. The main contents of the meeting recap is as follows:

I. Roles and Responsibilities of STEPI and VISTI

- The roles and responsibilities of STEPI and VISTI shall follow those mentioned in PCP, in principle.

II. Major Activities for 2019 and Actions Required

Kick-off Meeting and First Workshop

- The Kick-off Meeting and Workshop on National Future Strategy and Technology Foresight were conducted on April 15-19, 2019, in Hanoi, Vietnam.
- The summary of the major issues will be shared by STEPI and VISTI.

Capacity-building Workshop in Korea

- The Capacity-building Workshop will be held in late August.

➔ *Actions Required*)

- By mid-June, STEPI will share the draft of the program with the official letter and other technical issues (including information on the venue, number of invitees, and budget support)
- By the end of July, VISTI will confirm the list of participants and proceed with the necessary administrative process.

Draft implementation plan for the pilot technology foresight of Vietnam

- During the project period, STEPI will guide and review the plan for pilot study.
- VISTI shall select a specific sector/area/challenge that requires technology foresight; can be figured out throughout the seminar or workshop

➔ **Actions required)**

***By August and during the capacity-building workshop, VISTI shall suggest one or two national goals as candidate topics for technology foresight
During the period July-August, STEPI and VISTI shall confirm the direction of the pilot study.***

Final Workshop

- To be discussed later

III. Other Issues

Composition of the Project Team

- The internal team members of STEPI and VISTI have been composed.
- STEPI and VISTI will invite some external experts to be involved in the project.
- The honorarium for the advisory activities of the external experts will be paid by STEPI based on its administrative criteria.

➔ **Actions required)**

STEPI and VISTI shall share the list of external experts in Vietnam who are recommended by VISTI (and STEPI if applicable) ASAP

Information to be provided by VISTI

- VISTI's mandate (already delivered)
- Organizational structure of Vietnam's S&T Governance including important institutions
- To be added during the Project period

[Figure 4-1] Kick-off Meeting, Seminar, and Field Study



2. Capacity Development Workshop

2.1 Introduction

2.1.1 Objective

The workshop aims to enhance Vietnam's capacity in Science, Technology and Innovation (STI), especially focusing on technology foresight. The senior-level officials and managers of VISTI, relevant ministries, academe, and research institutions will be invited to participate in training programs. The lectures consist of various training courses from the introduction to technology foresight to hands-on project exercises.

2.1.2 Process

The workshop was supposed to be held in late August but has been postponed due to Vietnam's legal process.

During the period July to early August, STEPI and VISTI prepared for the workshop program and invitation of Vietnamese experts.

The following shows the program draft and the invitation letter sent by STEPI to VISTI:

[Table 4-5] Workshop Program (draft)

Date/Time		Venue	Description
Start	End		Activities and Time
DAY 1: Sunday, August 25, 2019			
		-	Arrival: HAN-ICN
DAY 2: Monday, August 26, 2019			
15:00	18:00		Orientation
18:00	19:00		Dinner
DAY 3: Tuesday, August 27, 2019			
09:00	12:00		Lecture 1 Introduction to the Korean STI System

Date/Time		Venue	Description
Start	End		Activities and Time
12:00	14:00		Lunch & Break
14:00	18:00		Lecture 2 Future Study and STI Strategy
18:00	19:00		Dinner
DAY 4: Wednesday, August 28, 2019			
09:00	11:00		Lecture 3 Introduction to Technology Foresight Korea's Experience: National Technology Foresight
11:00	13:00		Lecture 4 Introduction to Technology Foresight Pilot Project (hands-on project) Step 1: Environment Analysis (Value Chain Analysis & SWOT Analysis)
13:00	14:00		Lunch
14:00	18:00		Lecture 5 Step 2: Identify the List of Products with Priority
18:00	19:00		Dinner
DAY 5: Thursday, August 29, 2019			
09:00	13:00		Lecture 6 Step 3: Identify the List of Future Technologies Step 4: Analyze Future Technologies with the Delphi Technique
13:00	14:00		Lunch
14:00	18:00		Lecture 7 Step 4: Analyze Future Technologies with the Delphi Technique (continued) Wrap-up
18:00	19:00		Dinner
DAY 6: Friday, August 30, 2019			
09:00	18:00		Site Visit - KIST or KISTEP - Gwacheon National Science Museum
DAY 7: Saturday, August 31, 2019			
		-	Departure: ICN-HAN

[Figure 4-2] Invitation Letter from STEPI to VISTI



5th, 6th, 7th, Fl., Building B, Sejong National Research Complex, Sicheong-daero 370, Sejong 30147, Republic of Korea

Associate Prof. and Dr. Hoang Minh
President
Vietnam Institute of Science, Technology and Innovation
38 Ngo Quyen.
Hanoi, Vietnam

July 29, 2019

Your Excellency,

Subject: Invitation of the Vietnamese STI Delegation to the Republic of Korea

Science and Technology Policy Institute (STEPI) would like to express its warmest appreciation to the cooperation with your organization for the 2019 K-Innovation Project. For successful progress of the project, we cordially invite the Vietnamese Science, Technology and Innovation (STI) delegation of experts in STI policy and foresight to participate in the Capacity Building Workshop to be hosted by the Science and Technology Policy Institute (STEPI) in the Republic of Korea, from August 25th to August 31st, 2019. The tentative itinerary for the delegates is as follows:

- **Period:** Sunday, August 25 - Saturday, August 31, 2019.
- **Capacity Building Workshop:** Monday, August 26 - Friday, August 30, 2019
- **Funding to cover expenses related to the participation in the Capacity Building Workshop:**
From the 2019 K-Innovation ODA Program (Vietnam) - Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology and Innovation (STI) Strategic Planning in Vietnam 2019-2020. These expenses include only returned economic air ticket, accommodation, living expenses and local transport, but not per diem.
- **Delegates:** 10 member from Ministry of Science and Technology

This invitation and the attached program may serve as proper documentation to apply for visa at the Embassy of the Republic of Korea in Vietnam.

I and my colleagues of STEPI earnestly hope that your visit of this time will be very productive and meaningful. Please allow me to offer my best wishes for your enduring health and prosperity.

Yours sincerely,

Dr. Hwang-Hee Cho
President
Science and Technology Policy Institute
Cc: The Embassy of the Republic of Korea in Vietnam

2.1.3 Workshop Lecture Material

[Figure 4-3] Technology Foresight Manual

1. Introduction

1.1. Background

- There are significant uncertainties about the future shape of markets, governance and social values that will have an important impact on organizations and their capacity to meet their objectives
- The rapid technological and social change and uncertainties make the upsurge of interest in technology foresight, giving rise to its emergence as a global concept and policy tool
- Technology foresight can make unique contributions in the S&T policy to set research priority, to build networks, and to build visions of the future

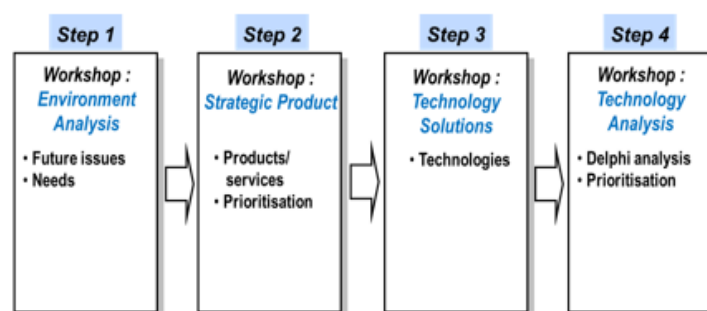
1.2. Purpose

- The purpose of this study is to provide the basic concepts of technology foresight and our experiences.
- This technology foresight study is expected to produce substantial references that can serve as a useful and practical guide to Vietnam when it plans to perform its national technology foresight exercise

1.3. Scope of the work

- This technology foresight study will be carried out in 4 work steps with the following scope:

- ✓ *Step 1 is intended to provide new perspectives about future, and identify issues and trends*
- ✓ *Step 2 is intended to identify STRATEGIC PRODUCTS which are considered to be important in the future of Vietnam*
- ✓ *Step 3 is intended to identify TECHNOLOGY SOLUTIONS in each sector and these will be target technologies for further analysis*
- ✓ *Step 4 is intended to analyze technology solutions in terms of their characteristics, importance, time of technology realization, and required government policies*



2. Environment Analysis

- Think of what issues/trends might be important in the future of Vietnam for the selected sector
- Collects and analyzes the trends/issues
 - Global trends/issues are gathered from literatures, web sites and by consultation
 - Identify the issues that may have an impact on the future using SWOT analysis for each activity of value chain in the workshop
- Identify key issues using cross impact analysis

(Ex)

□ *SWOT analysis for each step of value chain*

			Value chain				
			Inputs	Production	Processing	Storage/ Distribution	Markets
S W O T	Internal	Strengths					
		Weaknesses					
	External	Opportunities					
		Threats					

3. Strategic Products and Services

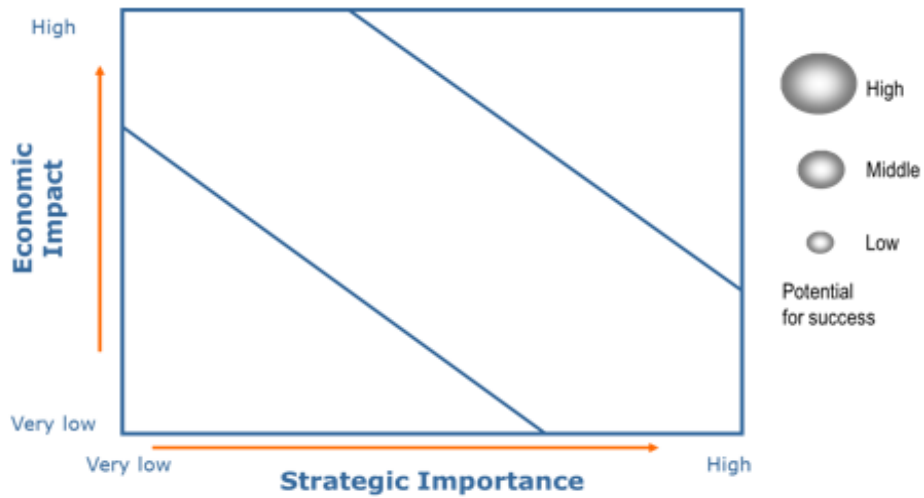
- Identify candidates of strategic products/services to respond to the key issues
- Identify strategic products/services with priority
- (Example) Participants are asked to draw criteria that are important to assess the candidates
- (Example) Participants are asked to assess the candidates in 5-point scale for parameters "economic impact", "strategic importance", and "potential for success"

(Ex)

☐ Identification of Products/Services

Value chain	Key issues	Related Actors	Products/Services
A.			
B.			
C.			
D.			
E.			

- *Prioritization to identify strategic products and services*

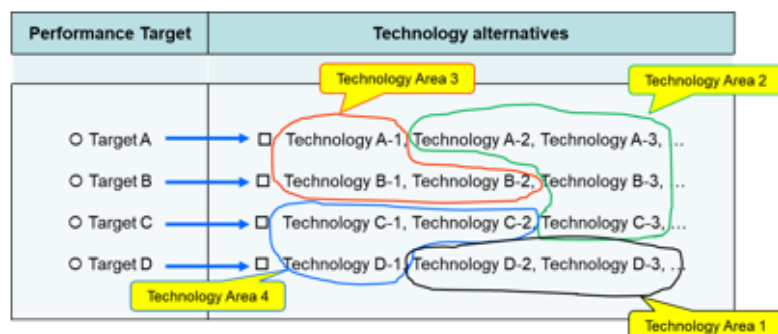


4. Technology Solutions

- Identify performance dimensions that are important for product performance, and that technology can deliver
- The 'Performance' provides a means of linking market, product and technology
- Explore the related technologies to meet the performance dimensions
- Arrange the technologies and develop the technology tree
- Similar technologies will be merged and technologies will be sorted into the appropriate groups

(Ex)

❑ Identification of Technology Alternatives



❑ Technology Tree

Tech Area	Tech Level 1	Tech Level 2



Figure 1. Example of Technology Tree by Component

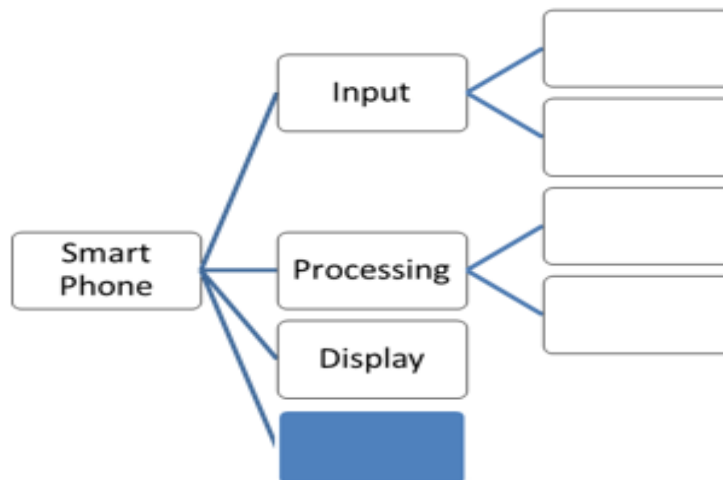


Figure 2. Example of Technology Tree by Functions

5. Technology Analysis

- Analyze technology solutions in terms of their characteristics, importance, time of technology realization, and required government policies



Figure 3. Example of technology analysis

- Each technology is analyzed by 2 round Delphi survey

What is Delphi?

- Delphi is a well-established expert-base technique which recognizes human judgement as legitimate and useful input for generating forecasts and exploring the future impacts of science and technology
- Single experts sometimes suffer biases while panels or group meetings suffer from 'follow the leader' tendencies and reluctance to discard previously stated opinions. These and other weaknesses in the use of expertise have led to the development of Delphi – a sophisticated survey aimed to collect and extract (tacit) knowledge from a group of experts by means of a series of questionnaires
- Questionnaires are normally carried out in rounds or waves (normally two, rarely three)

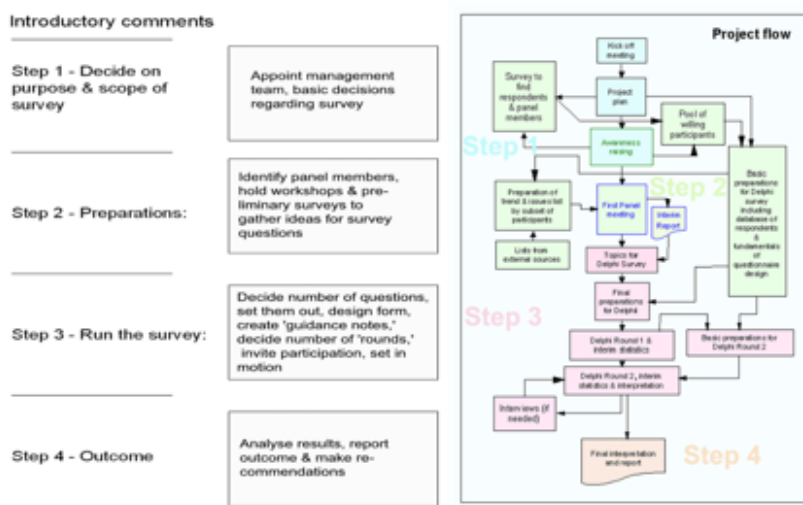


Figure 4. Delphi Process

- Questionnaires in the survey should be clearly and precisely phrased and parameters used in the survey should be comprehensively explained
- Identify key technologies

(Ex)

□ *Delphi survey questionnaire*

Technologies	Realization time				Significance in future society	Technical competitiveness
	'19-'24	'25-'30	'31-'36	>'36		

□ Identification of key technologies

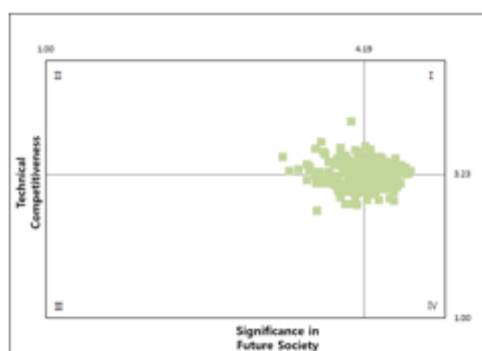
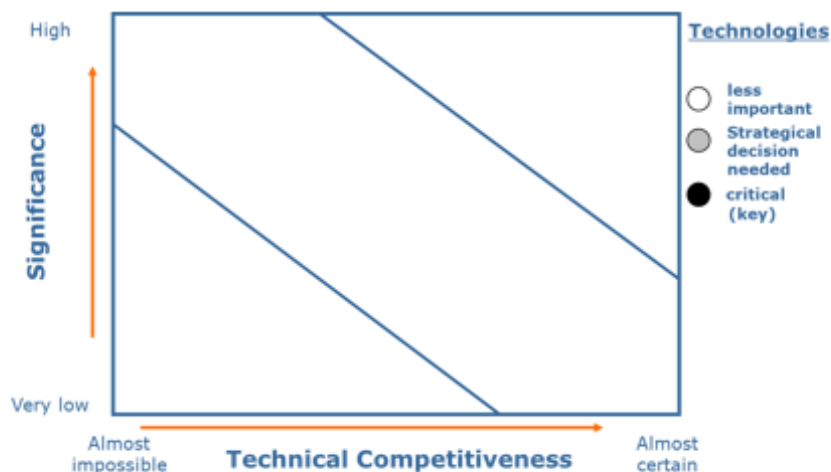
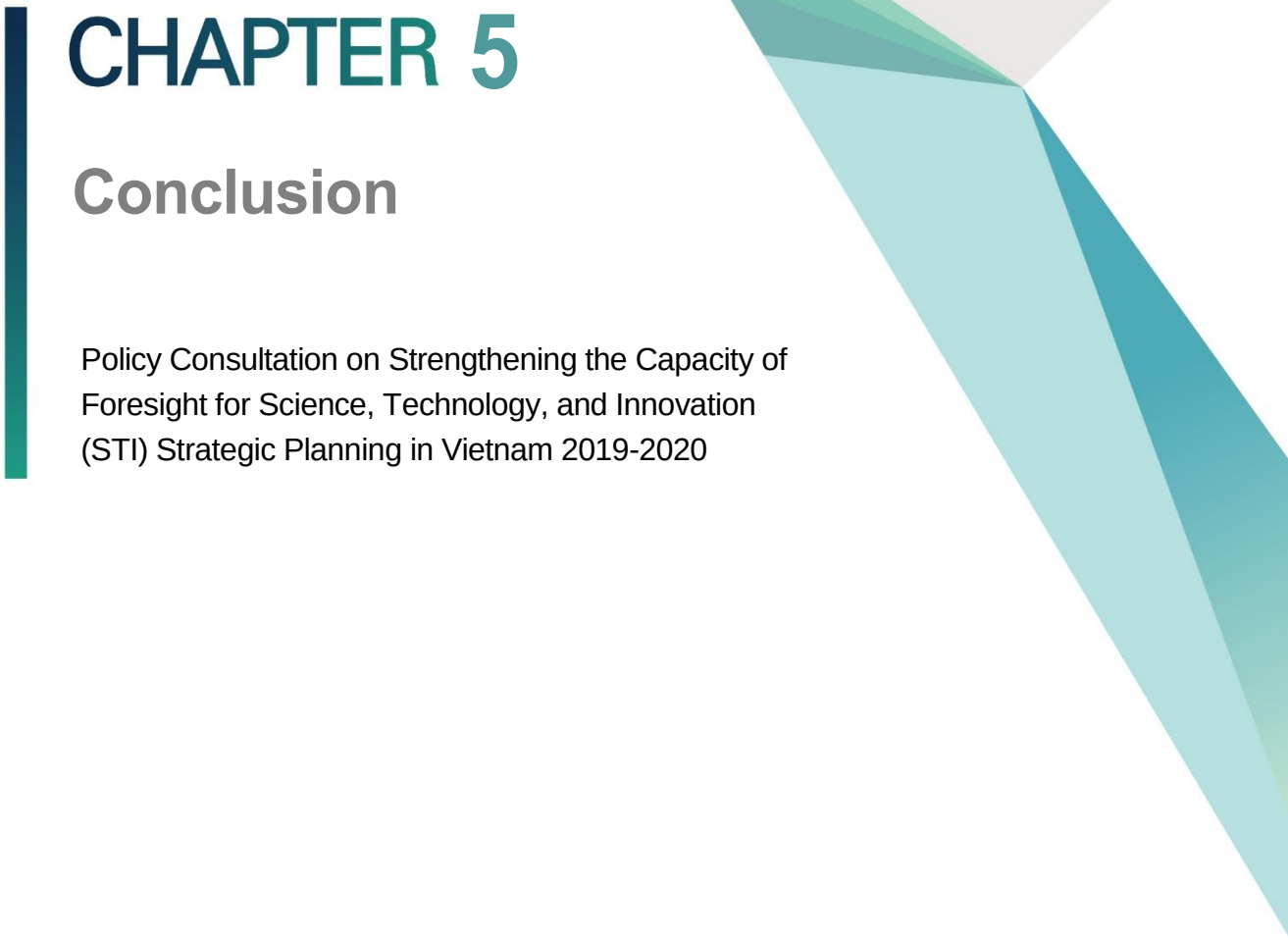


Figure 5. Dispersion of Significance and Technical Competitiveness of Future Technologies in the 5th Science and Technology Foresight

Classification	Group I	Group II
No. of Technologies	70	66
(Percentage)	(26.2)	(24.7)
Classification	Group III	Group IV
No. of Technologies	59	72
(Percentage)	(22.1)	(27.0)

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing left, and several overlapping triangles in various shades of teal and light blue, also pointing left.

CHAPTER 5

Conclusion

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation
(STI) Strategic Planning in Vietnam 2019-2020

Chapter 5. Conclusion

1. STI Policy and System

Vietnam's top national development plan is “Socioeconomic Development Strategy (SEDS) 2011-2020,” with “Socioeconomic Development Plan (SEDP) 2016-2020” established as an implementation plan to realize this. The main objectives here are to stabilize the macro economy and to increase economic growth, to restructure the economy and to change the growth model, and to enhance productivity and competitiveness. Vietnam has traditionally emphasized the importance of science and technology, and it considers science, technology, and innovation (STI) to be the most important pillar in the implementation of socioeconomic development strategies.

It is necessary to make the government establish and implement STI strategy systematically with active commitment to making STI contribute to national socioeconomic development. Note, however, that Vietnam does not have much experience in making and implementing such STI policies linked with the national development plan. Vietnam lacks linkage between national development goals – sectoral development goals – ministerial strategies, and action plans. In particular, this linkage should be clear to strengthen the role of STI. In order to establish the science and technology plan and strategy linked to national development goals and to take a new leap in line with the new economic and social changes, the STI system needs to be transformed. That means switching to a more creative, fast following system.

This requires an upgrade of Vietnam's science and technology innovation system. First of all, strategic investments and systematic management are needed to achieve the best possible outcomes given limited R&D resources. This means a simultaneous policy of the so-called

second-generation R&D and third-generation R&D in the research and development stage. The third-generation R&D system involves mapping the national development goals and science and technology goals and strengthening strategic linkages, whereas the second-generation R&D system includes systematically managing R&D activities through R&D evaluation, project management, and performance management. To this end, it should be reformed into evidence-based or data-based science and technology policy, which requires the establishment of a system for collecting, managing, and utilizing related data.

On the other hand, it is also necessary to consider improving the R&D structure to increase the performance of the R&D system given limited R&D resources. The role of R&D organizations such as VAST, VISTI, and VKIST, which are government or public research institutes, should be established. In addition, a linkage system with universities and industries needs to be strengthened for these public research institutes to create R&D synergy effects.

2. Technology Foresight

Technology foresight involves predicting and analyzing future technologies in all areas of science and technology, taking into account the changes in the internal and external environments. As suggested above, this is an important tool to link national development goals with science, technology, and innovation. In establishing and implementing a national socioeconomic development strategy, Vietnam can use the results of technology foresight to lay the groundwork for more strategic investments and policies. The national STI strategy should be linked to other sectors such as economy, society, and education according to the needs of the state, based on the results of technology foresight.

Vietnam has policy experience in performing technology forecasting or foresight, but such can hardly be seen as successful. One reason for this is the lack of experts in technology foresight and a very weak system of government cooperation. In addition, it is not possible to develop a technology foresight methodology suitable for the Vietnamese context by using the methodology of foreign countries without internalizing. It is important to develop Vietnam's own technology foresight methodology by applying the implications from the Korean experience to the Vietnamese environment and characteristics.

Using the results of technology foresights in STI policies and national development strategies requires clarifying the purpose at the planning stage. In recent years, the international community, including Korea, has placed the focus of technology foresight on the creation of innovation and the development of the entire economic society rather than science and technology development. Likewise, technology foresight is oriented toward a whole-of-the-government approach among ministries. In recent years, uncertainties have been increasing in society, such as politics, economy, and technology, all over the world, and the existing prediction methodology is confirmed not to be applied anymore. The new paradigm and necessity of foresight, rather than prediction, are highlighted anew. Therefore, technology forecasting and foresight are a process of finding direction, not content, with content to be filled afterward. In particular, Vietnam should explore its own growth path through technology foresight.

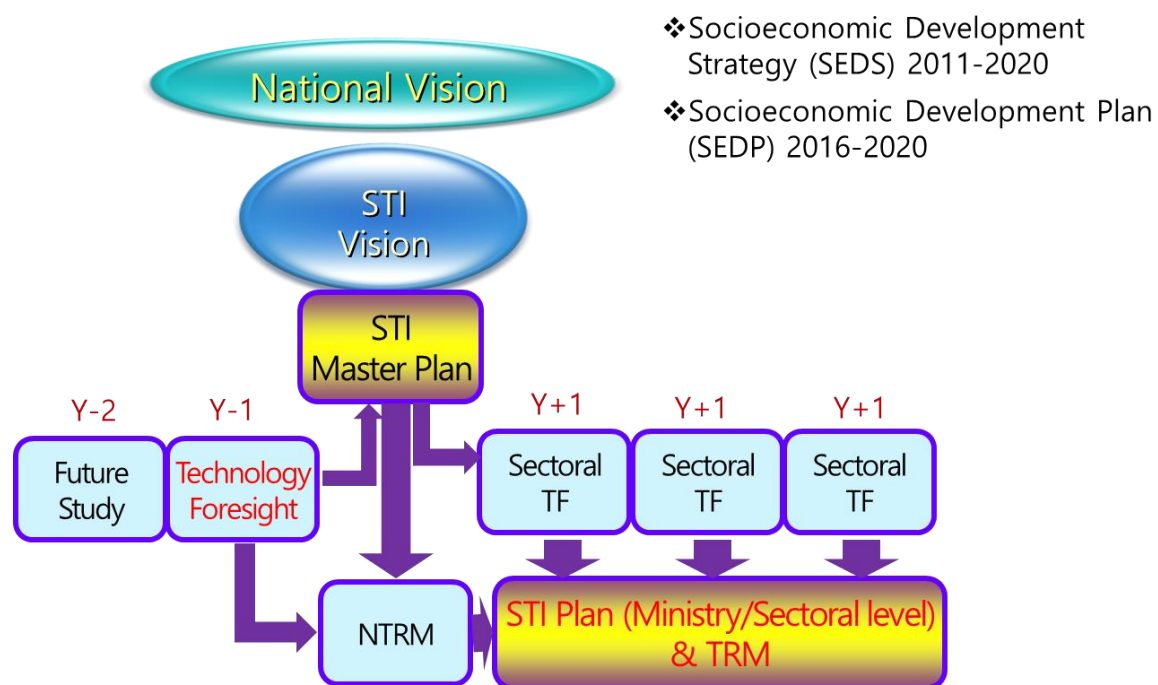
The roles and objectives of technology foresight also vary according to the stage of technology development. The role of scientists and engineers is important during the following phase, whereas the innovation ecosystem, or environmental infrastructure, is more important at the innovation phase. For the countries in the following phase, such as Vietnam, the search for technology solutions to current problems should be key elements of technology foresight.

Technology foresight can give the direction for developing countries like Vietnam to foster their core industries in the age of the Fourth Industrial Revolution. In particular, there is a high probability of success when appropriate and prompt support is provided through selection and concentration using limited science and technology resources. Based on the selection and concentration strategies, the STI policy needs several activities to be conducted with regard to technology demand surveys, technology forecasting, roadmaps, and analysis of STI capabilities.

The most important thing for the successful implementation and use of technology foresight is recognizing the importance of STI and technology foresight in the political decision-making process and cooperation of all the relevant government ministries. Extensive policy efforts are also needed to reconcile and coordinate the demands and interests of not only the government but also the people and social communities.

The STI plan and policy should be implemented in line with the national “Socioeconomic Development Strategy (SEDS) 2011-2020” and “Socioeconomic Development Plan (SEDP) 2016-2020.” For example, the government set the national future strategies based on future study. Technology foresight is followed to show the direction of the national STI master plan. The STI plan includes the vision, goals, and strategies of STI development as well as the key technologies needed to achieve the national development goals. For the next step, it needs a national technology roadmap for these key technologies. Finally, each ministry or institution in charge of each sector establishes and executes a strategy that includes a technology roadmap for each sector.

[Figure 5-1] Linking National Vision with Technology Foresight



Source: author

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